WONG YI LENG B. TECH. (HONS) BUSINESS INFORMATION SYSTEM SEPT 2017

DEVELOPMENT OF PARCEL2GO: AN INTEGRATED COURIER SERVICE APPLICATION

WONG YI LENG

BUSINESS INFORMATION SYSTEM
UNIVERSITI TEKNOLOGI PETRONAS
SEPTEMBER 2017

Development of Parcel2Go: An Integrated Courier Services Application

by

Wong Yi Leng
20239

Dissertation submitted in Partial Fulfilment of
the requirements for the
Bachelor of Technology (Hons)
Business Information System

SEPTEMBER 2017

Universiti Teknologi PETRONAS
Bandar Seri Iskandar
31750 Tronoh
Perak Darul Ridzuan

CERTIFICATION OF APPROVAL

Development of Parcel2Go: An Integrated Courier Services Application

by

Wong Yi Leng
20239

A project dissertation submitted to the
Business Information System Programme
Universiti Teknologi PETRONAS
in partial fulfilment of the requirement for the
BACHELOR OF TECHNOLOGY (Hons)
(BUSINESS INFORMATION SYSTEM)

Approved by,

_____

(Saipunidzam B Mahamad)

UNIVERSITI TEKNOLOGI PETRONAS
TRONOH, PERAK

SEPTEMBER 2017

CERTIFICATION OF ORIGINALITY

This is to certify that I am responsible for the work submitted in this project, that the original work is mine except as specified in the references and acknowledgements, and that the original work contained herein has not been undertaken or done by unspecified sources or person.

_____

WONG YI LENG

ACKNOWLEDGEMENT

First and foremost, I would like to take this opportunity to express my sincere gratitude to Mr Saipunidzam Mahamad for giving guidance and advice throughout this final year project period. I would also like to extend my gratitude to the users who willing to spend time to participate in the survey and testing by providing feedbacks. Furthermore, my grateful thanks are also extended to my friend for their significant help and advice given during the development of the system. Last but not least, I owe my profound gratitude to my parents for their support and encouragement during the development of this project. Without their support, advice and encouragement, this project would have not been successful.

ABSTRACT

There are numerous courier services available locally or worldwide. However, the most services only applicable at their respective company's whereby the pricing information is often hidden and customers usually need to call the company or visit their respective website or application to get quote, then make a price comparison with other company's courier services quotes for them to get their preferable quotation. This could be very time consuming mainly due to the inflexibility services whilst with certain terms and conditions apply only customers able to enjoy the door-to-door delivery service. Hence, the main objective of this project is to integrate a few of the courier services price and information in one stop whilst providing a low-cost peer-to-peer service to assist customer in delivering their parcel. The application will be able to generate the quotes of all the selected courier services in one application to ease the customer and enable a price comparison for each of the selected courier services company. The survey was conducted and analysis of respondents' feedbacks shows positive interest of having this application and would opt to use the low-cost peer-to-peer services. With Parcel2Go Mobile Application, the problem of inconveniency of the customer and the inflexibility of services that have in the current courier services will be solved as this application will ease the people who want to deliver their parcels.

TABLE OF CONTENT

CERTIFICATION OF APPROVAL ii

CERTIFICATION OF ORIGINALITY iii

ACKNOWLEGEMENT iv

ABSTRACT v

INTRODUCTION 1

1.1 Background of Study 1

1.2 Problem Statement 3

1.3 Objective 4

1.4 Scope of Study 4

1.5 Significance of Project 5

LITERATURE REVIEW 6

2.1 Courier Services 6

2.2 E-Commerce 8

2.3 Courier Services and E-Commerce Enterprise 10

2.4 Mobile Apps versus Mobile Website 11

2.5 Comparative Study 14

METHODOLOGY 16

3.1 Research Methodology 16

3.2 Project Phases 17

3.2.1 Initiation & Planning Phase 17

3.2.2 Analysis & Requirement Gathering 17

3.2.3 Design 18

3.2.4 Implementation 18

3.2.5 Testing Phase 19

3.3 System Design 19

3.3.1 System Architecture 19

3.3.2 Use Case Diagram 20

3.3.3 Activity Diagram 22

3.4 Key Milestone 25

3.5 Tools Required 25

RESULTS AND DISCUSSIONS 26

4.1 Preliminary Survey 26

4.1.1 Survey Setting 26
4.1.2 Pre-Acceptance Survey 26
4.2 System Interface Screenshots 29
4.3 User Accpetance Testing 33
CONCLUSION AND RECOMMENDATIONS 38
5.1 Conclusion 38
5.2 Recommendations 39
REFERENCES 40
APPENDICES 44
APPENDIX A: Gantt Chart 45
APPENDIX B: Preliminary Survey 46
APPENDIX C: User Accepting Testing 48
APPENDIX D: Parcel2Go Application Screenshot Interface 51

LIST OF FIGURES

Figure 3.1: Prototyping-Based RAD 17
Figure 3.2: System Architecture 20
Figure 3.3: Use Case Diagram 21
Figure 3.4: Activity Diagram 22
Figure 4.1: Pre-Acceptance Survey 27
Figure 4.2: Login Page 29
Figure 4.3: Get Quote Page 30
Figure 4.4: Delivery Request Confirmation Page 31
Figure 4.5: Delivery Status Page 32
Figure 4.6: Personnel Job Page 33
Figure 4.7: Customer's Feedback for User Acceptance Testing 34
Figure 4.8: Personnel's Feedback for User Acceptance Testing 36

LIST OF TABLES

Table 1.1: World Trade Organization…..……... 2

Table 2.1: Comparison of Similar System …….….. 14

Table 3.1: Key Milestone…….. 25

Table 3.2: Tools Required……..25

CHAPTER 1

INTRODUCTION

1.1 Background of Study

The population of Malaysia in the year between 2014 to 2016 has approximately 31.7 million and the annual growth rate between 2015-2016 is 1.5% (United Nations, Department of economic and social affairs). Among its evolving population, constant movement of people and goods from one location to another is expected to occur due to certain cause or situation (Prince & Bakon, 2016). According to Prince & Bakon (2016) agreed that it can be a quite stressful process for an individual to move from one house to another especially if the individual does not possess the resources (vehicle) to carry out such action. This barrier eventually occurs when deliver a parcel or goods to a particular location instigating that the individual will need to perform this action. Thus, door to door carriage services can be invaluable help for the individual (Agu, Nwoye & Ogbuokiri, 2015).

Courier services can be defined as an outsourcing medium whereby duty or function of transporting goods of a customer from one geo-location to another is responsible by a contracted company (Izzah, Rifai & Yao, 2016). Generally, it is related to delivery of parcels, packages, documents, letter and printed materials (Grunert & Sebastian, 2010) Typically, the postal transportation process consists of mail collection, input sorting, organize the movement of mail (global area transportation), output sorting and mail distribution. The complexity of the process thus it is better to be assisted by ICT to ensure that the correct mail can be received by the customer on time.

In Malaysia, the national postal services and private courier companies significantly contribute towards the national economy and also provide the employment opportunities. Due to the rapid evolution of Information Communication Technology (ICT), postal and courier services in Malaysia have undergone a radical change (Noordin, Hasnan & Osman, 2012). Despite of the advanced technology, the user who utilizes the services are still encountering the inconveniency of the end-to-end solution.

Courier Service	
Multi modal courier services	Services consisting of pick-up, transport and delivery services, whether for domestic or foreign destinations of letters, parcels and packages, rendered by courier using one or more modes of transport, other than by the national postal administration. These services can be provided by using either self-owned or public transport media.
Other courier services	Other courier services for goods, not elsewhere classified. For example, trucking or transfer services without storage, for freight.

Table 1.1: Source: World Trade Organization cited in (Noordin, Hasnan & Osman, 2012)

It also cannot be disputed that Booming Malaysia e-commerce industry to benefit from the delivery services. Wong (2016) stated that E-commerce has become a way of life for Malaysians, with 47% already using their smartphones to shop online. He also stated that approximately 7 million of them are already shopping online every month. Along with the fact that e-commerce industry expected to increase EUR 1 billion by 2020 in Malaysia and globally to US$1 trillion in the same years as stated by Najberg & Erickson (2015) thus all these businesses requires a high-quality logistics solutions to leverage this immense growth and meet the swiftly changing need of the online shoppers.

According to Ho (2015), the service experience which is the "O2O" model needs to be optimized whereby the transformation power lies in the offline service. Majority of the small start-up e-business may even need a last-minute delivery in order for them to complete the entire process. There are several businesses are promoting a 24-hour delivery and 4-hour delivery service has led to an intense competition within the E-commerce sector and among the stores. Subsequently, consumers will be more demanding for faster and better services. Ho (2015) also stated that consumer tend to switch brand easily with substitutes products available in the market which simply means that a comprehensive evolution is required for the order fulfilment delivery process. Nevertheless, the logistic service which is available in Malaysia is not flexible enough and may not be able to serve the customer's demands.

1.2 Problem Statement

An average person or online seller in the past could not easily access to the courier service providers. This is because majority of the well-established courier service providers require company registration, office address and a monthly committed volume for them to use their courier services. This has directly placed numerous online sellers in a dilemma situation as the high shipping cost of courier services would turn customers away, whereas normal mail services would take a very long time to deliver their packages to the buyers.

Besides that, customer find it time consuming to run around to quote rate. Customers often need to visit each of the courier service provider's website to get the best quote rate. At times, their websites do not even provide the quote rate and causing customer to give a call to the courier service company to obtain more information.

In addition, it is a quite stressful process for the customer to drop off their parcel at the nearest courier service offices if they do not possess any transportation of their own when they want to deliver their packages.

Furthermore, the rural regions face a particular challenge in the form of limited service area and relatively high cost of service delivery. This is because majority of

the prominent courier service providers mostly focus on the urban areas due to the lower density populations and larger distances that have to be travelled by service users and service providers in the rural area.

Therefore, the root cause found to this problem is the minimum used of service innovation with the available technology for the courier services to be done in more effective and efficient for way which can save half of the time for both average person and online seller.

1.3 Objective

- To integrate a few of domestic courier service providers quote rate in one stop to ease the users in terms of comparing the prices and information for each of the courier service providers.
- To develop a low-cost peer-2-peer courier service for matching up people who need parcels delivered and people willing to help them and earn while travelling.

1.4 Scope of Study

Based on the objective in the previous section, the scope of study will be only focusing on the major domestic courier service providers in Malaysia such as NationwideExpress, PosLaju, CityLink and GDex. This mobile application will only develop for the Android phones. It is also mainly focusing on how this application is able to save the courier service user's time in terms of getting instant quote rate from different courier service providers. In addition, courier service users could enjoy the pick-up delivery service without any registration process whilst this application could help those who willing to help and earn while travelling.

Besides, the scope of study will also be focusing on assisting the courier service users who wants to deliver their parcels to the rural areas which prominent courier service providers do not cover the rural areas. Therefore, this application will be focusing on two categories whereby the first category is by integrating the major

courier major domestic courier service provider’s quote rate and information whilst the second category whereby a door to door carrier service is available.

1.5 Significance of Project

Based on the problem statement as mentioned, this integrated courier service application can make a significant impact on the courier service users by saving up their time in terms of getting quotes from different courier service providers. Besides, it could provide an alternative solution in covering the rural areas which is not covered by prominent courier services. Furthermore, a pick-up delivery service will be provided for people or an online seller who are busy and inconvenient to travel without any registration process. Thus, with this Parcel2Go mobile application, it could provide an alternative to the inflexibility of courier services.

CHAPTER 2

LITERATURE REVIEW

2.1 Courier Services

Courier services can be defined as a contracted company whose responsibility as an outsourcing medium in assisting the role of transporting the goods of a customer including documents, parcels or other packages from one geo-location to another. According to Reviews (2016), a courier company is one which delivers messages, packages and mail is recognized got their speed, security, tracking service and specialization. The process of collecting mail at the collection centre, sorting the mail base on location delivery destination, the movement of mail to ensure timely delivery of mail and distribute the mail to the delivery destination is included in the process of postal and courier services (Izzah, Rifai, & Liu Yao, 2016). Today, with the advancement of technology by ensuring that the correct mail could be received by the customer on time, the Information Communication Technology (ICT) is implemented in the process of postal and courier services assist the efficiency of mail delivery process.

Courier services operate on all scales, which are from town to town, district delivery services and worldwide service. Among the courier companies consist of DHL, GDEX, UPS, FEDEX and etc. Izzah, Rifai, & Liu Yao (2016) stated that the delivery pricing must be competitive and flexible where it is also easy to use and reasonable rates to the value of goods, weight of goods, in addition the distance of delivery. A right pricing plan shall be automatically provided to the confidence of the customer (Lin, Choy, Ho & Lam, 2014). Therefore, courier services today could be also defined in such a way that the trust between courier services as a business

partner that logistic services e- commerce by improvising the performance of e-business from the business side with delivering important parcel.

Courier service is considered as the one of the service sector that is boosting the economy in Malaysia. Malaysia's participation in Trans-Pacific Partnership Agreement (TPPA) is likely to boost the courier service sector by 20% within the next 5 years. (Ninikas, Athanasopoulos, Zeimpekis & Minis, 2014)

Nevertheless, the result of globalization has proven that this industry encounter challenges in managing the creativity in services. The existence and growth of logistic management companies faces numerous challenges ahead in terms of the ability of creativity innovative services, accessing to expert capital, and the services and operation even if there is a 7% annually of growth in average in the courier Market of Malaysia. the European Courier, Express and Parcel (CEP) could be taken as an example for a very tactical and dynamic sector in the globalization.

In order for Malaysia to develop post ally and courier service management more competitively, Malaysia still needs to do a continuous learning and research. For instance, Kunkel, Doppstadt & Schwind (2010) mentioned that the European Courier, Express and Parcel (CEP) sector is a very dynamic and strategic sector involved in globalization and the CEP sector primarily is an activity which is separated from the postal service. According to Grunert & Sebastian (2010), Malaysia courier services today is to understand the type of products that are profitable in addition not handling with the continuous modification in the nature of business based on the e-commerce environment irrespective of their scopes deal with challenges in facing with the technology advancement from a business partner. Hence, a constant change is needed to be made to gain competitive advantage in e-commerce business environment, services and operations, and customer where the only determinant of the courier service industry is no longer only the price (Khan, Dominic & Naseebullah, 2010)

2.2 E-Commerce

Electronic commerce is a type of business model, or segment of a larger business model, or that allows a firm or individual to conduct business over an electronic network, typically the internet. Basically, it operates in all four of the major market segments which consist of business to business, business to consumer, consumer to consumer, consumer to business.

Buying and selling which is considered as a new interactive market has been created with the advent of the internet as an open network with the utilization by millions of people online. Therefore, Jehangir, Dominic, Naseebullah & Khan (2011) agreed that the activity of e-commerce basically includes the capability of buying and selling items in addition where the public network allows in delivering marketing information and online-services.

The rapid growth of E-commerce can ensure the smooth operations on business transactions via online networks. Kenyon & Meixell (2011) mentioned that the smoothness of operations has led to the increase in service quality, lower cost in business operations, customer satisfaction, faster and more efficiency transaction including the delivery of physical goods by Logistics Service Providers and quick delivery of goods in some cases such as software or digital music.

On December 2014, BBC news reported that Malaysia only commence to use the Internet as the core platform at around 1995. High-speed internet access and its accessibility and affordability is provided by the government to the Malaysians until the year of 2010 has provided by the government. In the year 2015 State of Internet, Malaysia was ranked 70th place internationally however still considered as the country fall behind in the widely usage of internet.

Malaysia still regards the commercialization of the Internet as a new platform. This is because majority of them have had the perceptions that the usage of internet is only cater for the communications and entertainment industry and not acting as a medium for commerce. According to Kenyon & Meixell (2011), e-commerce in Malaysia is still in the early stage in the development as to compare to the western countries, conducting transactions online is not that popular. Although the Internet

users in Malaysia stated that they prefer the concept of online shopping but the fact that only a minority of them performing the online transaction.

However, from the year of 2010 to 2015, the speed of internet which is getting more advance and affordable causing Malaysia began to feel the reliability in using the internet for e-commerce application in addition to the current trend regardless of the age level. (Khan, Dominic & Naseebullah, 2010).

The enterprises of e-commerce allow consumers to shop electronically where it could be performed at anytime and anywhere without restrictions of time or distance. E-commerce has been rapidly growing in the last five years and in subsequent years; a superior performance with quicker pace is expected to be shown.

B2B refers to a type of transaction that exist between businesses, for instance one involving a manufacturer and wholesaler, or a wholesaler and a retailer which often include the logistics service providers as logistic provider.

The e-commerce businesses usually are dependent on the logistic service providers for completing their operation and management. Therefore, Izzah, Rifai & Yao (2016) has explored the e-commerce company's activities which requires the logistic service provider to provide them the logistic management and services in the warehouse management or to facilitate the process of delivering goods. E-commerce is a platform that could benefit the business operations in various areas. The mentioned benefits include a convenient, cheaper and faster than the conventional approaches of battering the goods and services.

E-commerce provides great advantages in business operations such as cheaper convenient and faster than the conventional methods of battering goods and services. It had led the development of the electronic market place where customer and suppliers conduct mutually beneficial trade together. Therefore, it is very vital for the company to minimize the gap between what the company can do about the idea of e-commerce and what company should do as an online business in order to create a successful online store.

2.3 Courier Services and E-Commerce Enterprise

In Malaysia, the logistic service is basically equivalent to the courier service companies. It is mutual that the courier services companies often being address as a Courier Partner Logistics (CPL) or business partner. Generally, CPL provides any logistics services to cater their business operations.

According to Izzah, Rifai & Yao (2016), the relationship between both the courier / logistic services companies have a symbiosis relations where the courier service providers and customer (e-commerce enterprise) will use the logistic service repetitively. The logistic management strive to boost competitive enhancement by carefully assimilating the core functions inside the organization to other business. Whilst with the outside operations of supplier activities, customer prerequisite and other network members should be connected effectively with the business partner. To express it in another term, the outcome of the customer perspectives will be resulting a good organizational performance with business partners like courier services companies as a CPL to e-commerce enterprise (Sterba & MacCallum, 2010).

Their business partner enterprise could only be enhanced with the condition of the capability in providing the best services to them and making them feel satisfied and confident. Therefore, performance of the company could be enhanced together where they have a great performance in the relationship to supply chain operations.

Kersten & Koch (2010) mentioned that logistics are not limited to manufacturing operations solely and it is applicable to all types of organization for instance to serve e-commerce enterprises Certainly, all the enterprises particularly in the retail industry, large organization, small and medium enterprises could relate to the logistic management (Xu & Wang, 2010).

Apart from that, Izzah, Rifai & Liu (2016) stated that the collaboration in the courier service offers multi-innovative features and require a long time to build a friendly alliance in courier service and business partner. Hence, the need of flexibility in services to meet the wants and demands of e-commerce enterprise companies is vital for the courier service providers. From here we can observe that how the performances of e-commerce enterprises could positively be affected by the courier services.

New demands in the courier services have been formed due to the rapid development of e-commerce enterprise in network. For instance, e-business's customer who has demanded the different which simultaneously provide a service which is value-added in courier services are increasing. Formerly, the conventional method of its standing made it hard to seek for services to meet these new needs. Thus, the development of e-commerce enterprise which can improvise the position of courier service providers in the forthcoming as relevance is analysed, where it is also impactful for the economy development in Malaysia especially.

2.4 Mobile Apps versus Mobile Website

In the era technology of the desktop PC the web browser was the main method of access to these resources; however, everything has changed due to the explosion of smartphones and tablets. In spite of every handset being equipped with a mobile browser, users are instead choosing to use apps for various common functions such as downloading and consuming music and videos, and reading news (Borley,2014).

Native apps basically are downloaded applications- from Apple's App Store, Android Apps on Google Play. They could not be accessed within a browser and only could be installed on a mobile device. Although it is a separate entity form a company website, it often supplements the brand in certain ways. Apps are allowed to provide service to individuals through a smaller platform rather than wanting them to go through the website (Blue Fountain Media, 2015)

The world of apps is also seeing a rise in popularity with smartphone penetration currently at 50 percent in the U.S. (A global study of consumers' expectations and Experiences of Mobile Application) A Nielsen survey emphasized that the trend that found American smartphone users spend four times as many minutes each month using mobile apps as they do their mobile web browser back in 2012 (Colwyn, 2014). According to Badalian (2015), customers are likely preferring native apps. This is because millennial want convenient, functional and relevant mobile app experience (Bhagat2015).

Badalian (2015) also mentioned that companies who put out native apps will be gaining an instant competitive advantage as compared to organizations which are opting for mobile web and getting bitterly disappointed with the response of their customers. This could be proven with the case of Delta's native Fly Delta App which got the award of best achievement in technology for the Airline Passenger Experience Association (APEX) in year 2012 and its superior performance versus its competitor's mobile web apps (Staff Writer, 2017). Customers who fly regularly mostly is relying on user-friendly, native function to quickly obtain information, submit itinerary changes, check in, or perform other actions. Whilst mobile web simply do not provide the smooth experience where native apps are capable for.

On the other hand, mobile website is similar to any other websites whereby it consists of browser-based HTML pages that linked together and accessed over the Internet. The noticeable characteristic that distinct a mobile website from a standard website is the fact that it is designed for smaller handheld display and touch-screen interface (Summerfield, 2017)

According to Angeles (2014), the prime benefit is that the mobile users are able to access to the regular website. It could obtain all the same elements as what they contain in the regular version of the website for instance its design and content of the pages and it features a mobile friendly layout that offers improved readability and functionality when viewed on a smartphone or tablet. Angeles (2014) also further explained that customers can access to the website anytime, anywhere using any mobile device without conceding the user experience by having a mobile website.

Despite of the benefits of mobile website, users today largely still prefer mobile apps as it provide a better control of display. Borley (2014) stated that displaying information using the same visual interface across all devices is undesirable where the largest tablets will have smaller screens than the average laptop PC. He agreed that the using an app could become much easier to shape and define how information is displayed on screen.

Besides that, mobile apps could have a better control of navigation. According to Borley (2014), the apps provide much more granular control of the user experience regardless of the improvements in mobile user navigation. This is because of the

buttons and links can be sized for “finger-friendliness”, reducing the risk of unwanted link clicks inherent with re-sized websites on mobile – fat finger syndrome.

Lastly, mobile apps encounter a lesser chance of clicking-out. By assisting user to keep engaging with the information to be presented and reduce the high chances of being distracted available to user, the isolationist approach is used. Whist websites have always been open to lost traffic as users click links leading off site and never to go back (Borley, 2014)

Hence, we can see that mobile apps today is the current trend now where majority of the user will seek to opt for mobile apps rather than mobile website because of its capability in providing a better user experience as to compare with mobile websites since apps provide “one-click access” unlike a mobile website that requires to take several clicks and text entry to access.

2.5 Comparative Study

In the course of this research, it is discovered that there are other applications are similar to the proposed system. There are quite a number of similar applications exist in the Google Play and AppStore but only three standard applications namely ezyCourier, FastFast and eCourierz has been selected for review in this research. Although these applications have similarities, they do not have a functionality whereby both the price comparison of domestic courier services and a peer-to-peer delivery service is also provided in one-stop application. Table 2.1 illustrates the comparison made between the several mobile apps.

Features / Application	**ezyCourier**	**FastFast Delivery**	**eCourierz**	**Parcel2Go (Proposed System)**
Type & Quantity Sending	• Document – up to 500g • Box – up to 5kg • Luggage – up to 20kg	• Document (Up to 35x27cm) • Small Parcel (Up to 25x 20 x10cm) • Medium Parcel (Up to 35x32x 10cm) • Large Parcel •	• Documents up to 2 kg • Parcel- Up to 5kg	• Document – up to 2kg • Parcel – Up to 30kg
GPS Tracking of Goods	Could track goods in real time	Could track goods in real time	It is still not available in the apps yet.	Could not track goods in real time

Payment Method	Cash on Delivery; Online (Credit Card)	Cash on Delivery	Cash on Delivery; Online Payment	Cash on Delivery
Place of Coverage	Malaysia, China, Vietnam and Canada	Singapore	India	Klang Valley (Malaysia)
Google Maps	Built in-apps. Convenient for user to provide accurate address	Built in-apps. Convenient for user to provide accurate address	Built in-apps. Convenient for user to provide accurate address	Fill in pick up and drop off location
Delivery cost	Clear and instant pricing	Clear and instant pricing	Only Courier services price is shown	Clear and instant pricing
Delivery Status	No pop-up notification to notify	No pop-up notification to notify	No pop-up notification to notify	There is a pop-up notification to notify
Availability	Depends on the availability of the personnel	Hundreds of drivers on-call, 24 hours a day	Only working hours	Hundreds of drivers on-call, 24 hours a day
Method of Delivery	Door-to-Door delivery service	Door-to-Door delivery service	Door-to-Door delivery service is not available	Door-to-Door delivery service
Personnel Service Rating	Available for user to provide feedback on the provided personnel	Not available for user to provide feedback on the provided personnel	It is still not available yet	It is still not available yet
Price Comparison with a few courier services	Not available. Could not compare which is the best price and timing of delivery	Not available. Could not compare which is the best price and timing of delivery	It is available. Could compare price rate	It is available. Able to view and compare the best price
User Interface	Not user friendly. All details cram into a page.	User friendly. Flow of ordering is easy	Not user friendly. Certain module is directing to the mobile website	User friendly. Flow of viewing price rate is easy and flow of order is easy too.

Table 2.1: Comparison of Similar System

CHAPTER 3

METHODOLOGY

3.1 Research Methodology

The Rapid Application Development (RAD) based methodology has been selected as the most suitable methodology for the Parcel2Go Android Application project because it is a small-to-medium scale project and only short duration of development has been given which is approximately 4 months.

Rapid Application Development (RAD) is a methodology that provides a faster development and higher quality system as compared to the conventional life cycle. This methodology was chosen because it is designed in such a way that it could take advantage of the powerful development software for instance CASE tools, prototyping tools and the code generator to speed up the analysis, design and implementation phases in order to get certain portion of the Parcel2Go Android Application developed rapidly and into the hands of the users for evaluation and feedback.

The feedback from the users will then reanalyze, redesign, and re-implement a second version of prototype that corrects deficiencies and add more features. This methodology is very useful when the Parcel2Go Android Application users have difficulty in expressing their requirements for the system because it could rapidly provide a system for users to evaluate and reassures users that progress is being made.

Hence, prototyping-based RAD is the most suitable methodology for requirements of the system which is uncertain because they are able to provide the users to co-operate in the early stage of the development and a short time frame given to develop

the android application. The final outcome of the android application is targeting on providing the ease of use for the users who wants to compare the prices of different domestic courier services and meanwhile could opt for a low-cost peer 2 peer delivery.

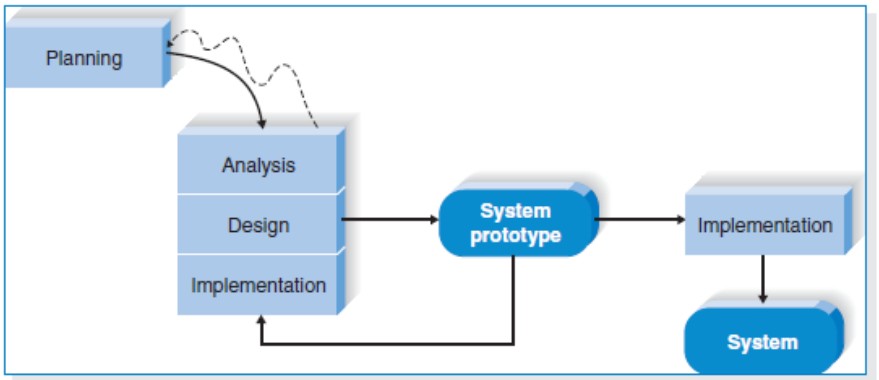


Figure 3.1: Prototyping - Based RAD

3.2 Project Phases

3.2.1 Initiation & Planning Phase

The problem for this project has been identified through the experience of the e-commerce and an average person who wants to send a parcel to another party. After conducting certain research, an android application is needed to integrate the price rating of certain domestic courier services in one stop and meanwhile provide a low-cost peer-2-peer courier service to ease both online seller and an average person.

This project is mainly divided into two phases. Information gathering and research will be conducted in the first phase in the period of 11 weeks whilst the application development will be conducted in the second phase in the period of 14 weeks. Thus, total of 25 weeks is required to complete this project development. A work plan which includes the project's Gantt chart will be generated at the end of this phase as a guideline to perform the tasks that scheduled to develop the system as shown in Appendix A.

3.2.2 Analysis & Requirement Gathering

A critical analysis on the literature is conducted in this phase is to obtain a depth understanding on the current challenges faced by the online seller and an average

person who wants to deliver parcels or documents and also to review the relationship of the e-commerce and courier services company. A comparative study of the similar existing system with the proposed system is also conducted in this phase. Subsequently, a set of close ended questionnaires will be distributed to gather user requirements about the proposed system – Parcel2Go Android Application and will then be analysed and determine how their requirements fit the proposed system.

3.2.3 Design

Unified Modelling Language (UML) such as Use Case Diagram and Activity Diagram will be designed in this phase to ensure the user requirements are fulfilled whilst modelling application. The system interface and database of the proposed system will be designed to enable user to have a clearer picture of the system. The proposed system architecture will also be designed to determine the two-way communication between the customers and our parcel2Go worker with the web server and database. The system architecture, UML Diagram and system interface will be discussed in next topic.

3.2.4 Implementation

In this phase, the development of the prototype and user's evaluation will be conducted concurrently. The first version of prototype will be developed and presented to the user with questionnaires. From the feedbacks from the user, evaluation will be made based on the functional requirement and subsequently the prototype will be refined. The refined prototype will then be again presents to the users and evaluated to produce the final product of the prototype until the user requirements are met. The final outcome of this phase will be the documentation of the adjustment of final version of the prototype.

3.2.5 Testing Phase

In this phase, test requirements will be made and documented base on the last version of the prototype. Unit testing and functionality testing of Parcel2Go Application with be conducted in this phase to ensure that there are no errors in the system. After that, a user acceptance test will then be carried out to see if the system is accepted and then a final documentation of the completion of the Parcel2Go Android Application will be the outcome of this project development.

3.3 System Design

3.3.1 System Architecture

The design of the system architecture for the integrated courier service application is shown in Figure 3.2. From the illustration of the Figure 3.2, there will be a two-way communication which involves both parties which is the customer and the Parcel2Go personnel. Whenever customer wants to deliver their parcel, they will first need to send a delivery request and the order information will be process through the webserver and store in the database before the Parcel2Go personnel could receive the customer's order. After the personnel had received the order from customer, the personnel will need to give a confirmation to the customer. Confirmation is then again go through back to the web server and stored in confirmation into the database before customer received the confirmation. There are two levels of user authentications which have a different role of permission in accessing the contents and functionalities. For instance, the customer is able to access the content of sending a booking order request whilst the Parcel2Go personnel have the functionality of accepting or rejecting a customer's request.

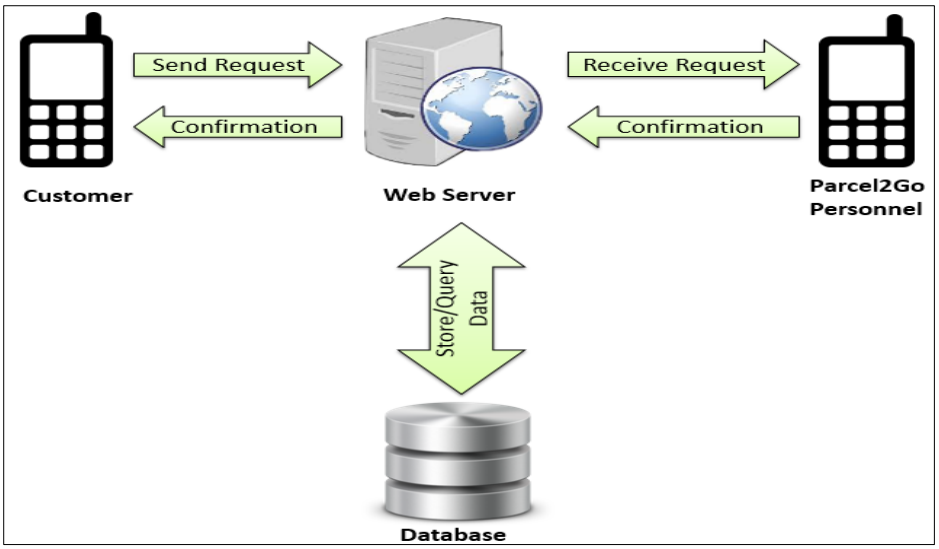


Figure 3.2: System Architecture

3.3.2 Use Case Diagram

Figure 3.3 shows the use case diagram for this integrated courier service application. Three different actors engage in this system which is the customer, Parcel2Go personnel, and the system admin.

Customer could login to the mobile application using their registered email address. The application system will then verify identification of the customer. After successfully login to the application, the application will direct the customer to the main page of the application. The customer need to create a new delivery request by entering the item information and delivery destination. Subsequently, the customer able to view the quote rate of different courier service providers.

Customers who prefer the Parcel2Go services could accept the delivery service and view their delivery information in the confirmation page. The OTP code is able to be viewed upon the successful delivery request that has been made. Apart from that, customer can view their item delivery status and personnel information. Lastly, customer could logout their account if they do not want to use.

As for the Parcel2Go personnel use case, the personnel have the same ability as the customer by logging in and out from the application. The personnel could view a list

of pending delivery request and to select and accept the delivery request. After accepting the request, the personnel could view the customer details and update both the delivery status

From the Figure 3.3, the system administrator could calculate the delivery rate of few courier service providers and display out it out for the customer to view. System admin also can instantly update the delivery status of the item right after the personnel update to the system.



Figure 3.3: Use Case Diagram

3.3.3 Activity Diagram

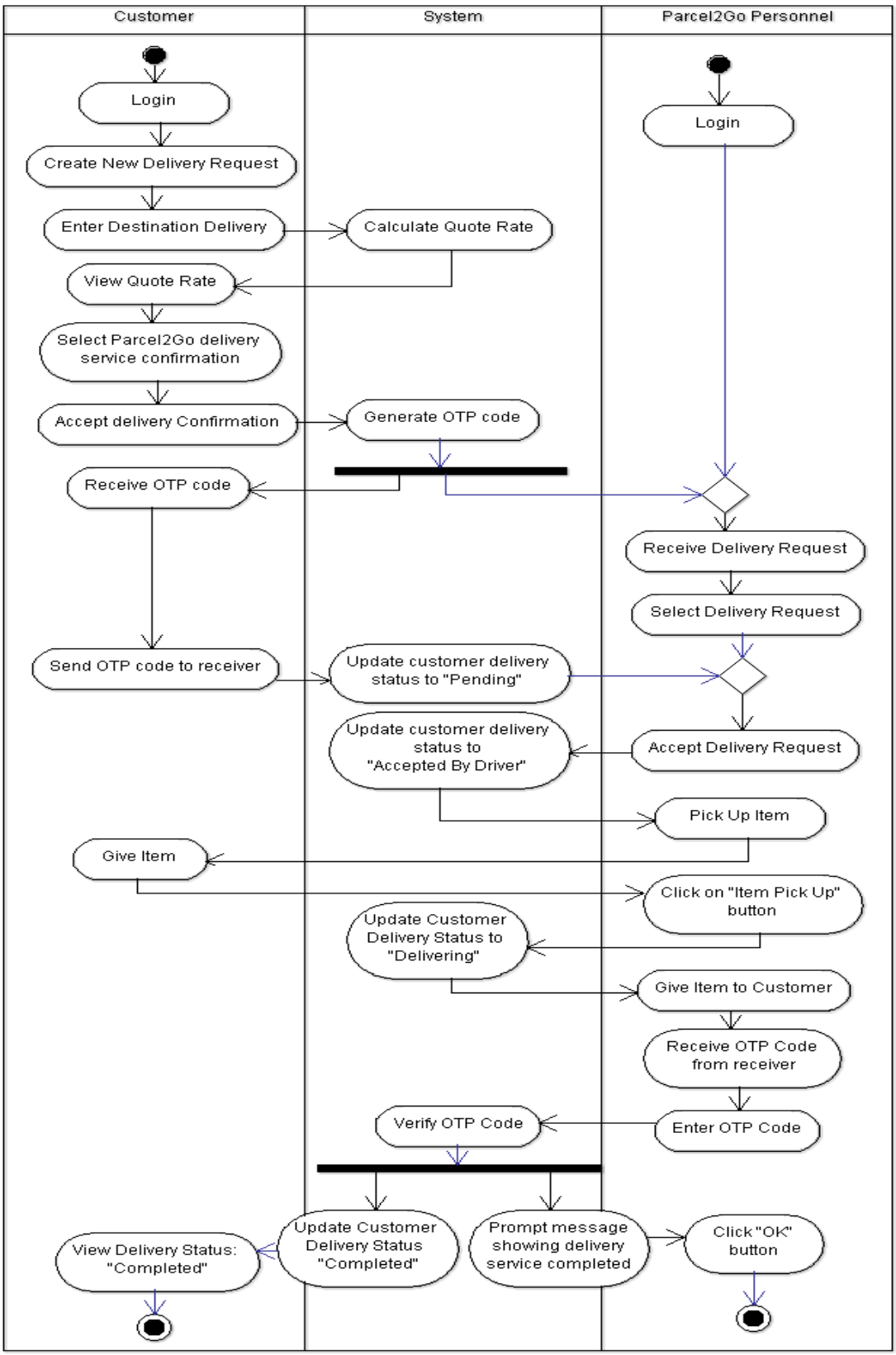


Figure 3.4: Activity Diagram

Figure 3.4 depicts the activity diagram of the Parcel2Go Application. It shows the flow of the application. The application involves three parties such as customer, system and Parcel2Go personnel. Both the customer and personnel will need to use their registered email address to successfully access to the application. After the customer has login to the application, the customer is required to create a new delivery request. After that, they need to enter their delivery destination and the system will help to calculate the quote rate based on the details keyed in. Subsequently, the customer is able to view a list of quote rate from different courier service provider. If the customer has chosen the Parcel2Go Delivery service, they could view and accept their delivery confirmation and subsequently the system will generate the One Time Pass (OTP) code for the customer. The customer will need to send the OTP code to the receiver. The system then now will update the customer delivery status to "Pending". The personnel now could receive the delivery request and choose whether to accept the delivery request. If the personnel have accepted the request deliver, the system will now then update the customer delivery status to "Accepted By Driver". Once the personnel click on the "Item Pick Up' button after picking up item from the sender, the system will then again update the delivery status to 'Delivering". After the personnel has delivered the item to the receiver, it will then need to request OTP code from the receiver to be enter into the receiver address page. The system subsequently will verify the OTP code and update the delivery status to "Completed" whilst it could be view by the customer. Meanwhile for the personnel side, it will also prompt a notification showing that the delivery service has been completed and personnel will just need to click on "OK" button. The whole application will end with the flow.

3.4 Methods of Data Collection

For this project, data is collected using the following data collection methods:

(i) Quantitative Research

A quantitative research on this mobile application has been conducted by using the Google Forms. The online survey has been conducted involved a large sample size of respondents whose aged between 18 and 45 years who use the courier service frequently. The respondents came from various backgrounds for instance an online business seller, teachers, housewives, students and other professionals. The purpose of this quantitative survey is to determine the current practice or traditional process of people and courier service providers. By having this quantitative method of data collection, a numerical result of user acceptance regarding this mobile application can be gathered and ready to be use and implemented in the application development. *(Refer to Appendix B)*

(ii) Pre-Acceptance Survey

A set of close ended questions are forwarded to the respondents which is a frequent user of courier service and those who have experience in wanting their parcels to be delivered. In addition, the respondents also include those who are willing to earn while travelling. The survey has been distributed to 50 respondents from various backgrounds whose aged between 18 and 45 years and the result will be discussed in next chapter. The survey is intended to study the user's acceptance of having an integrated courier service application to make their booking delivery a breeze.

(iii) User Acceptance Testing

Reviews and feedbacks based on testing of the prototype of the project will be collected from the courier service users to verify whether the features and functionality of the mobile application can suit the needs of users. 60 respondents from various background such as students, teachers, professionals, full time housewives, and online seller has tested the

system prototype. The system prototype involves two different type of users such as the customers and personnel. Both the users were asked to complete an online survey regarding their experience after using this application. The results of the user acceptance testing will be discussed in the next chapter.

3.4 Key Milestone

Table 3.1 has displayed the key milestone that needed to be achieved throughout the project of Parcel2Go Android Application.

Milestone	Estimated Date
Project Approval	3/6/2017
Results and Discussions	30/6/2017
UML Diagram Design	12/7/2017
Present First Prototype & Questionnaire to User	30/8/2017
Present Refine Prototype & Questionnaire to User	20/9/2017
Document Adjustment Made in Final Product	10/10/2017
User Acceptance Test	25/10/2017
Final Documentation of the Completion System	3/11/2017

Table 3.1: Key Milestone

3.5 Tools Required

Tools that are required are divided into two categories which are the tools for research and tools for application development.

Research Tools	Application Development Tools
Laptop	Android Studio 2.1
Online Articles	Android SDK
Google Scholar	Firewall Database
Google Play	Android Device (Table/Mobile Phone)
Journal	Android Device Manager

Table 3.2: Tools Required

CHAPTER 4

RESULTS AND DISCUSSIONS

4.1 Preliminary Survey

4.1.1 Survey Setting

Despite the fact there is a similar existing application that already exist in market, however there are still certain inflexibility in the existing application whereby the online business seller and courier service users find the inconveniency in them when comes to delivering parcels. Therefore, development of Parcel2Go that has been proposed is to improvise the existing apps in such a way that the dilemma and the inconveniency that the courier services users are facing could be solved.

A preliminary survey has been conducted with users who often have parcels that need to be delivered and with those who have experienced in delivering their parcels. A set of close ended questionnaires have been distributed to the related respondents. The Likert Scale is a five-point scale where 1 signifies strongly disagree by the users and 5 signifies strongly agree by the users. The survey that has been conducted involved 50 respondents whose aged between 18 and 45 years who will deliver their parcel using the courier services. Among the 50 respondents, they were from various backgrounds including online business sellers, teachers, students, fulltime housewives, other professions etc.

4.1.2 Pre-Acceptance Survey

Throughout the course of survey, it has been found out that majority of the courier service users in Malaysia are still facing the inconveniency of delivering items. Regardless of the fact that courier service providers today have offered various

solution to meet their customer needs, there are still certain inflexibility of courier service providers causing the users face inconveniency.

The results of the preliminary survey have been discussed as shown in Figure 4.1.

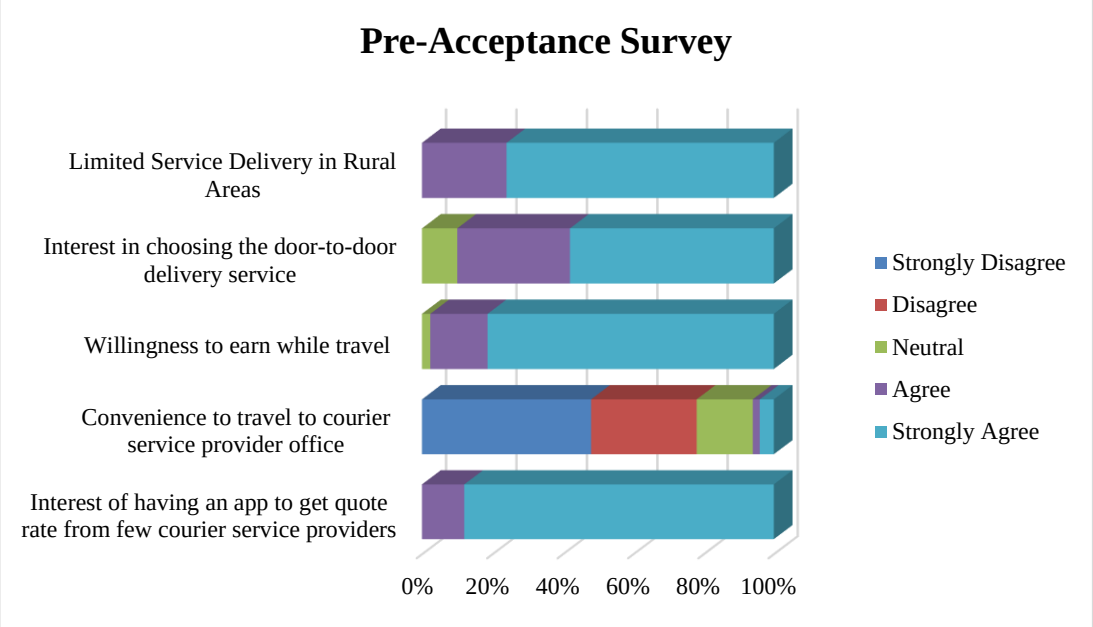


Figure 4.1: Pre-Acceptance Survey

According to the Figure 4.1, it illustrates that 76% and 24% of the respondents strongly agree and agree respectively that the well-established courier service providers in Malaysia has a limited service delivery in rural areas. This scenario has caused the need of customer to travel to the nearest branch to either drop-off or receive their parcels. This is because most of the courier service providers mainly focus on the urban areas as rural regions have a lower density populations, longer distances that have to be travelled by service users and service providers, and also the small number of people in any location that preclude economies of scale.

Besides that, 58% of the respondents strongly agree that they have interest in choosing the door-to-door delivery service to help them to deliver their parcel. This is because this service could help people who are very busy and could not leave their workplace to drop off their parcel at the nearest post office. In addition, 32% of the respondents agree with the service provided whereby they could skip the long queues during peak hours in post office and only 10% of the respondents remain neutral. Thus, the respondents were very interested in having such services provided without any lengthy process needed to be done as compared to the well-established courier

service providers which requires customer to register a few documents first only able to enjoy the door-to-door delivery service.

Furthermore, Figure 4.1 also demonstrates that the willingness of people to earn while travel from one location to another consist of 70%. They are very interested in the concept of earning side income by helping people to deliver parcel according to their convenience as anyone can be the personnel and no rigid employment working hour that they need to abide to. If the respondents possess their own vehicle with a clear record, then they could assist people in delivering parcels. Conversely, only 2% of the respondents remain neutral. This most likely that the respondents like the idea of earning while travel but does not possess their own transportation thus remain neutral.

Moreover, 48% of the respondents strongly disagree with the statement that it is very convenient to travel to the nearest courier service provider office to drop off their parcel. The reason is that most of them does not possess their own transportation and is a stressful process for them to travel from one location to another. In contrast, only 2% of the respondents agree that it is convenient for them to travel to the nearest post office because it is most likely that they possess their own transportation and it would not be a concern for them to travel anywhere whenever they want to. Therefore, majority of the respondents strongly agree and show a keen interest by having a door-to-door delivery service to solve their inconvenience travelling to the nearest courier service provider office.

Lastly, the figure 4.1 also displays that the respondents who show a strong interest of having an application to get instant quote rate from a few different courier service providers consist of 88%. Whilst 12% of the respondents also agree that by integrating a few courier service provider's quotations in one stop can save their time in terms of price comparison. This is because people no need to run around to quote rate. Courier service users also find it cumbersome when they need to go to each of the courier service provider's website to get quote rate or sometimes they might even need to give a call to the customer hotline service to get clarification for further information about quote rate. Thus, with this Parcel2Go application, it could solve the inflexibility of current courier service providers whilst provide alternative for the users.

4.2 System Interface Screenshots

The designed system interface, features and functionality of the integrated courier service application has been developed based on the data gathering analysis by illustrating the idea of application. The following figures portrays the interface screenshots of the developed application.

4.2.1 Login Page

There are two types of user for this application such as customer and personnel. The system will prompt the same login page for both users as shown in Figure 4.2. The customer and personnel will have different level view of access by differentiating them with their registered email address after successfully login.

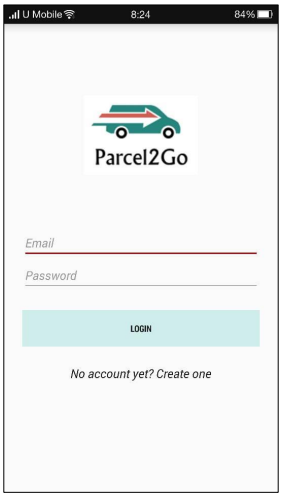
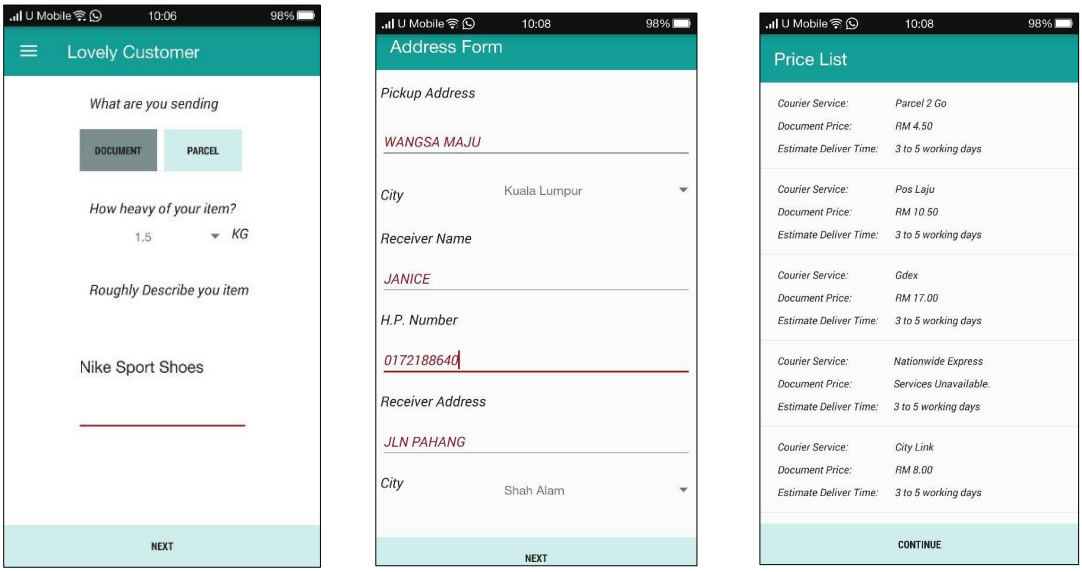


Figure 4.2: Login Page

4.2.2 Get Quote Rate Page

After customer has successfully login to the system, the system will firstly prompt the customer to enter the type, weight and a brief description of the item that needed to be delivered as illustrated in Figure 4.3(a). Next, the system will direct the customer to the address form page which demonstrated in Figure 4.3(b). Customer must enter their pick-up address and receiver's information include their name, contact number and address. By proceeding to following page shown in Figure 4.3(c),

the system will assist the customer in getting the instant quote rate from Parcel2Go service itself and four different major domestic courier service providers in Malaysia such as PosLaju, GDex, Nationwide Express and CityLink.

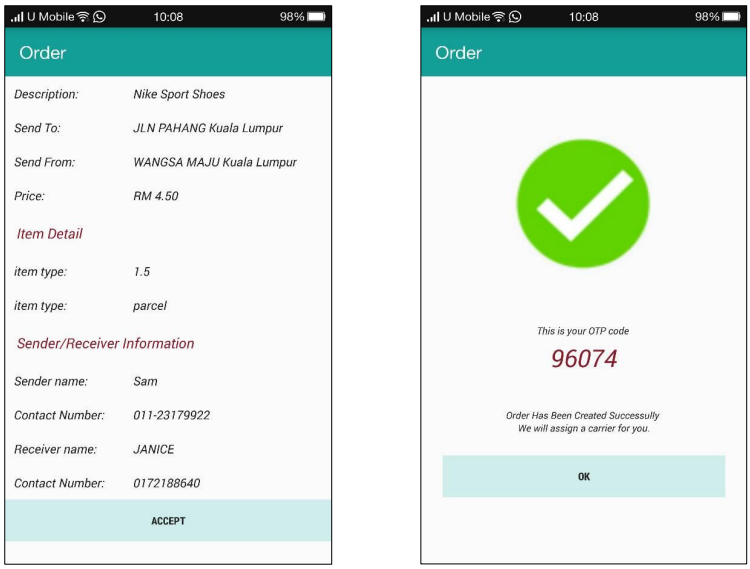


(a) (b) (c)

Figure 4.3: Get Quote Page

4.2.3 Delivery Request Confirmation Page

If customer choose to use the Parcel2Go delivery service to help them to deliver their items after viewing their quotation, customer could proceed to the following page illustrated in Figure 4.4(a) by confirming their delivery details that they have entered earlier. Subsequently, Figure (b) displays a message indicating that order have been successfully made will be generated along with One-Time Pass (OTP). The customer will then need to send the OTP to the receiver.

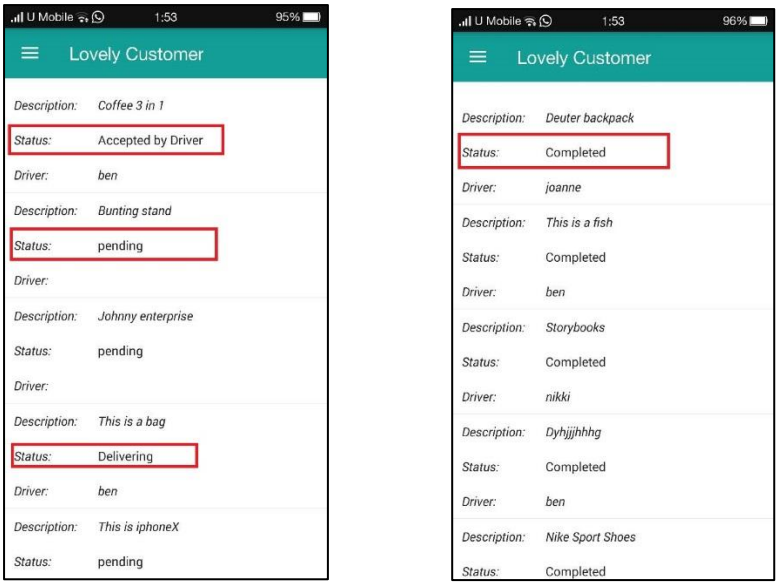


(a) (b)

Figure 4.4: Delivery Request Confirmation Page

4.2.4 Delivery Status Page

Based on the Figure 4.5(a), it has illustrated four types of status delivery. There are mainly “Pending”, “Accepted by Driver”, “Delivering” and “Completed”. When a delivery request has been successfully made, the status of the delivery will remain “Pending” until the Parcel2Go personnel have accepted their delivery request and then only the status will change to “Accepted by Driver” in addition the name of driver will be appeared. Once the personnel had picked up the item from the customer, their delivery status will change to “Delivering”. Later, when the receiver has received their item and give the OTP code to the personnel to key it in to the system, the customer delivery status will automatically change to “Completed” as display in Figure 4.5(b).



(a) (b)

Figure 4.5: Delivery Status Page

4.2.5 Personnel Job Page

The system will direct the Parcel2Go personnel to a list of pending delivery request as shown in Figure 4.6(a). The personnel could choose their preference of delivery service from the list of pending delivery request. Subsequently, a page with the details of the selected customer delivery service will be display for personnel to view as demonstrated in Figure 4.6(b). Next, after accepting the delivery request, a pop-up notification message of confimration will be prompt for personnel to make sure its confirmation as depicted in Figure 4.6(c). After accepting, a sender's details page shown in Figure 4.6(d) is direct by the system for the personnel to go and pick up the item. Personnel could update the system by clicking on the "Item Pick Up" button and the system will switch to the receiver's details page shown in Figure 4.6(e). The personnel need to request the OTP code from the reciver to be entered into the system. Lastly, once the system indicates that the OTP code is successfully verified, it will pop-up a notification showing that "You just completed a delivery service" as demonstrate in Figure 4.6(f).

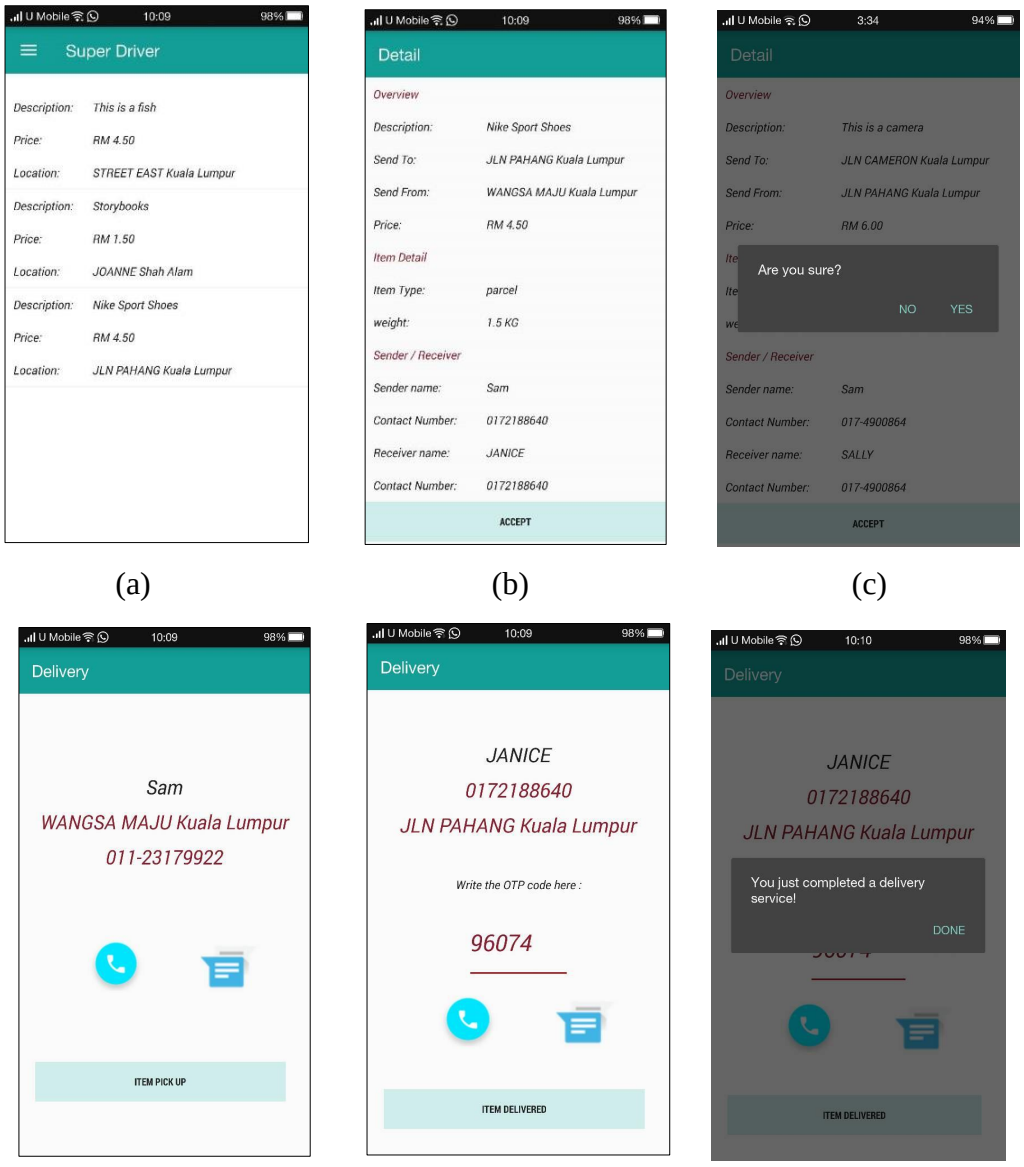


Figure 4.6: Personnel Job Page

4.3 User Accpetance Testing

The usability testing has been conducted to 60 respondents of users of the integrated courier service application. The respondents are from various backgrounds such as student, online business seller, professionals, full-time housewives etc. All the respondents were given a mobile phone that has been installed the Parcel2Go Mobile Application. There are 2 types of user for this mobile application such as customer and personnel. Both the users were asked to complete an online survey regarding their experience after using this application. The survey consists of 5 questions in the

feedback form which basically covers the interest of the customer who are the courier service users, user friendliness of the application and to determine the usefulness of the application.

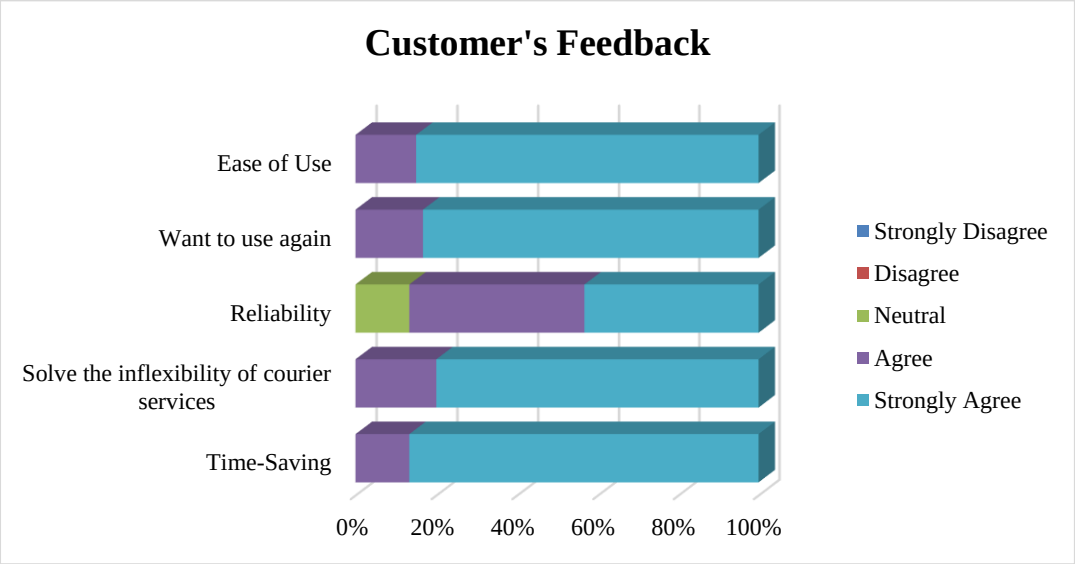


Figure 4.7: Customer's Feedback for User Acceptance Testing

The results of bar chart in Figure 4.7 describes that 85% of the respondents strongly agree that the application is easy to use. The mobile application developed in such a way that it is very straight forward. Despite that some of them who is a non-tech savvy person, he or she is still able to use the application.In conjuction to that, the reaminng 15% of the respondents also agree that this mobile application is as ease of use. Throughout the course of user acceptance testing, respondents gave a positive feedback on the ease of use application.

In addition, 83.3% of the respondents strongly agree that they want to use the application again because they found that the concept of service innovation in this integrated couroer service application makes booking delivery a breeze. If they have parcel that need to be delivered, they can just get everything done at their comfort zone without the need of purpose travel to the nearest courier service provider's office. The remaining 16.70% of the respondents also shows the interest of using this mobile application again as it could assist customer in getting quoatation from the different famous courier service providers.

From the Figure 4.7, the respondents for both strongly agree and agree in terms of the reliability of the system comprise the same which is 43.30%. The respondents found it reliable as the application also could view their delivery status. The application will update the customer's delivery status each time whenever their request is accepted by the personnel, collected the items and received by the receiveer. The customer will be able to know who is coming to help them to deliver their parcel. Meanwhile, only 13.30% of them remains neutral on the reliability of the system.

Based on the Figure 4.7, we could also analyze that 80% and 20% of the respondents both demonstrated that they strongly agree and agree that the concept of this developed mobile application could solve the inflexibility of the current courier service providers. The inflexibility of courier service providers here oftenly refers to the registrations process that need to done by the customers in order for them to enjoy the dor-to-door delivery service. In addition, this mobile application provides an alternative for the customers who want to deliver their parcel to the very rural area whereby most of the courier service providers mainly focus in providing their delivery service in the urban areas.

Lastly, 86.70% of the respondents in Figure 4.7 strongly agree that the Parcel2Go mobile application can save up their time. In additon, 13.30% of them also agree that on the similar scenario. They agree that this applicatioin save up their time in terms of getting quote rate from different courier service providers. They do not need to run around to get their prefereable rate as it will be time consuming. At times, the quote rate that provided in their respective website also might find it confusing and causing the respondents to give a call to their office to further clarify about the quotations. Thus, by having this application, the system will help you to get an instant quote rate based on the weight and distance of the item.

Apart form the customer's feedback, the personnel which also involves in the system application are also required to conduct a user accpetance testing and provide feedbacks regarding the application. The personnel's feedback results are as shown in Figure 4.8.

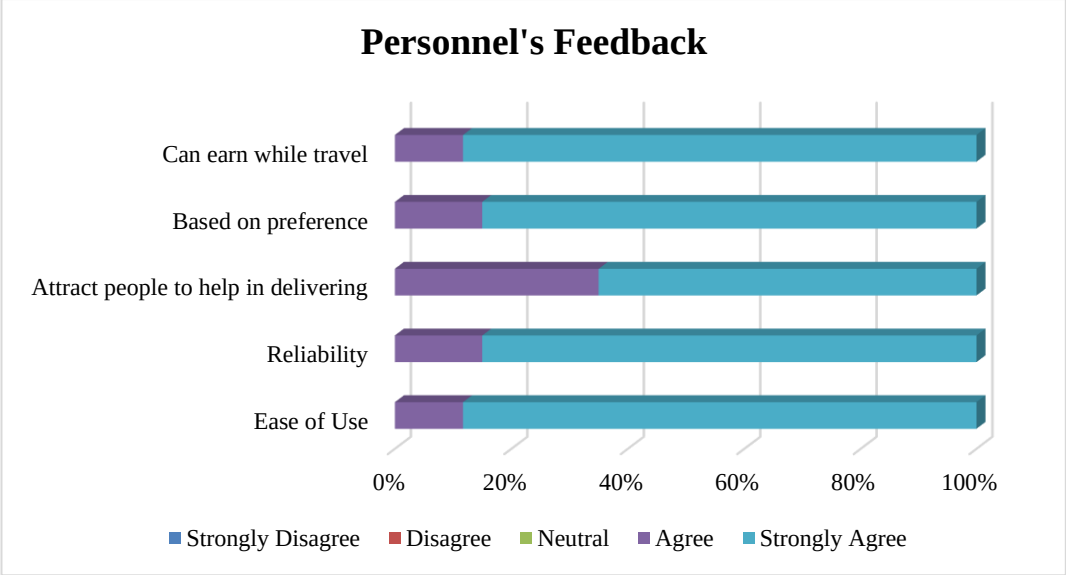


Figure 4.8: Personnel's Feedback for User Acceptance Testing

Based on the bar chart in Figure 4.8, 88.30% of the respondents strongly agree with the concept of the system application that they could earn extra while travel by helping people to deliver parcels. In conjunction to that, the remianing 11.70% of the respondents also agree that they can earn while travelling. Majority of the respondents like the concept of earning extra while travel because they could go according to the flexibility of their timing. Anyone also could be the personnel of Parcel2Go as long as they possess their own transportation with clear records.

In addition, the respondents who strongly aree and agree that they could go according to their preferrence in terms of their convenience in accepting the delivery request from customer consists of 85% and 15% respectively. For instance, if the personnel's current location is not far apart from the delivery request destination, personnel could opt to pick up and drop off the item which is as requested.

With this concept of service innovation in Parcel2Go, 65% of the respondents strongly agree that this concept could attract more people to help in delivering as the younger generation today like to earn according to their preference and flexibility whereby there is no rigid employment hour that they need to abide too.There is a nil percentage of the respondents disagree that the Parcel2Go service innovation that can attract people to help in delivering.

Figure 4.8 also illustrated the reliability of the system application has been strongly agreed by 85% of the respondents and agree by 15% of respondents. This is because once after they have accpeted the customer delivery request, the customer side will able to see that their delivery requested have been accepted along with the personnel's name generated. After accpeting the request, whenever the personnel update the system with the customer's item delivery status, the customer is able to view. The personnel will need to ask the One Time Passcode from the receiver in order for the system to update that their delivery service has completed and notify the customer at the same time. Hence, the system has proven the reliability in terms of its delivery status.

Lastly, 88.3% of the respondents strongly agree with the ease of use of application and nil percentage of them strongly disagree that the system is not in ease of use. The reason is that the whole system interface is developed very straight forward and goes accordingly with the flow. It will not cause the personnels to confuse with the flow of application to ease their delivery process. Personel who is not a tech-savvy person could also able to catchup quickly with the whole process of system application.

CHAPTER 5

CONCLUSION AND RECOMMENDATIONS

5.1 Conclusion

To conclude, in the era of increasing rapid technology advancement have assist the courier service providers in providing a more efficient and effective in the process of delivery. Today, the courier service application is widely used. The usage of these applications has become important for people who frequently want their items to be delivered, especially for people who are conducting online businesses.

According to the preliminary survey conducted, most of the courier service users find it time consuming and difficult getting quotation from the different courier service providers in addition to the inflexibility in their service delivery. The inflexibility of courier service includes the limited service providing to the very rural areas and a lengthy registration process required to enjoy the door-to-door delivery service.

Hence, the integrated courier service application that implement the concept of service innovation to solve inconvenience of courier service users in getting quotation and the stressful process of travelling to the nearest courier service provider's office to drop off the parcel. With the Parcel2Go Mobile Application, the system could generate the quotation from different courier services whilst matching up people who needs parcels to deliver from their doorstep to another and people willing to help them and earn while travelling.

This could be proven with the results of testing indicates that this concept of service innovation in Parcel2Go has benefit the courier service users by providing an alternative solution to them in the process of delivering their parcels.

5.2 Recommendations

There are still room of improvements for this mobile application to be made in the future in order to make it more reliable and user friendly for the courier service users by helping them to deliver their parcels with just a one-click access with a preferable rate and suggestion. There are few features and aspects that could be consider for future enhancements in this mobile application: -

(a) Implementing a Google Map in to the application to ease the user in entering their delivery destination
(b) Implement a real-time tracking delivery to increase the security and trustworthy of the customer.

All in all, the significant of this project is to improve the concept of service innovation in the inflexibility of courier service providers today whereby it could benefit the users and the society in terms of parcel delivery.

REFERENCES

Agu, M.N., Nwoye, C.I. and Ogbuokiri, B.O (2015) 'Enhancing Courier Service with the Development of an Interactive Mobile App in Android Platform', IOSR J. Mobile Comput.Appl, (IOSR-JMCA).e-ISSN, pp.2394-0050.

Angeles, S. (2014, July 15). Mobile Website vs. Mobile App: What's the Difference? Retrieved 29 June 2017, from: http://www.businessnewsdaily.com/6783-mobile-website-vs-mobile-app.html

Badalian, J. (2015, 20 May).HTML5 vs Native: The Debate is Over.Retrieved 29 June 2017, from:https://www.mobilesmith.com/html5-vs-native-debate-is-over/

Bhagat, R. (2015). Oracle's latest report sheds light on how millennials use mobile apps. Retrieved 29 June 2017, from: https://e27.co/millennials-are-finicky-a-poor-mobile-experience-can-cost-you-a-customer-20150514/

Borley,R. (2014, 27 May). Why smartphone apps are killing the mobile web browser. Retrieved 29 June 2017, from: http://dootrix.com/smartphone-apps-killing-mobile-web-browser/

Colwyn,S. (2014 13 March). New consumer media consumption research. Retrieved 29 June 2017, from: http://www.smartinsights.com/marketplace-analysis/customer-analysis/consumer-media-device-use/

Dennis A., Wixom B., and Roth R., (2006).System Analysis and Design John Wiley and Sons, Incpg 17.

Grünert, T., & Sebastian, H. J. (2010).Planning models for long-haul operations of postal and express shipment companies. European Journal of Operational Research, 122(2), 289-309.

Ho. M (2015). New On-Demand Delivery Service to power the growth engine of Malaysia e-commerce in 2015.Retrieved 1st July 2017, from: https://www.linkedin.com/pulse/new-on-demand-delivery-service-power-growth-engine-malaysia-maggi-ho

Jehangir, M., Dominic, P. D. D., Naseebullah, N., & Khan, A. (2011). Towards digital economy: the development of ICT and e-commerce in Malaysia. Modern Applied Science, 5(2), 171.

Kersten, W., & Koch, J. (2010).The effect of quality management on the service quality and business success of logistics service providers. International Journal of Quality & Reliability Management, 27(2), 185-200.

Kenyon, G. N., & Meixell, M. J. (2011). Success factors and cost management strategies for logistics outsourcing. Journal of Management and Marketing Research, 7, 1.

Khan, M. J., Dominic, P. D. D., & Khan, A. (2010, June). Adoption of e-commerce in Malaysia and its effect on the business performance: An organizational perspective. In Information Technology (ITSim), 2010 International Symposium in (Vol. 1, pp. 1-5).IEEE.

Kunkel, M., Doppstadt, C., &Schwind, M. (2010). Open service-oriented computing for logistics: A case in courier, express and parcel networks. In Service-Oriented Computing. ICSOC/ServiceWave 2009 Workshops (pp. 134-144). Springer Berlin/Heidelberg.

Lin, C., Choy, K. L., Ho, G. T., Lam, H. Y. (2014).A decision support system for optimizing dynamic courier routing operations. Expert Systems with Applications, 41(15), 6917-6933.

MuhamadAriff, N.R. and Davies, H. (2009) 'Sustainable living environment for urban low-income households in Malaysia: key factors for maintenance', In

CIBW107 2009: Proceedings of the 2009 International Symposium On Construction in Developing Economies: Commonalities among diversities (pp. 380-393). CIBW.

Najberg A. and Erickson J. (2015, June) Cross Border E-Commerce to Reach $ 1 trillion in 2020. Retrieved 1st July 2017, from http://www.alizila.com/cross-border-e-commerce-to-reach-1-trillion-in-2020/

Ninikas, G., Athanasopoulos, T., Zeimpekis, V., & Minis, I. (2014).Integrated planning in hybrid courier operations. The International Journal of Logistics Management, 25(3), 611-634.

Noordin, A., Hasnan, N., & Osman, H. (2012). Service Innovation of Postal and Courier Services in Malaysia: Will It Lead to Customer Responsiveness. International Proceedings of Economics Development & Research, 42, 205-209.

Prince,E.B. &Bakon, K.A. (2016). A Study on The Need to Implement a Courier Service Application on Android Smartphones. International Journal of Information System and Engineering , 4(2).

Reviews, C.T.I. (2016).Media of Mass Communication. (8 ed.). Cram101 Textbook Reviews, 2016.

Staff Writer (2017, Feb 16). Delta: One of world's most-awarded airlines. Retrieved 29 June 2017, from: http://news.delta.com/delta-one-worlds-most-awarded-airlines

Sterba, S. K., &MacCallum, R. C. (2010). Variability in parameter estimates and model fit across repeated allocations of items to parcels. Multivariate Behavioral Research, 45(2), 322-358.

Summerfield,J. (2017, N.D.). What's the Difference Between a Mobile Website and an App (Application)? Retrieved 29 June 2017, from: https://www.hswsolutions.com/services/mobile-web-development/mobile-website-vs-apps/

United Nations, Department of Economic and Social Affairs, Population Division. World Population Prospects: The 2015 Revision. Available at http://www.un.org/en/development/desa/publications/world-population-prospects-2015-revision.html

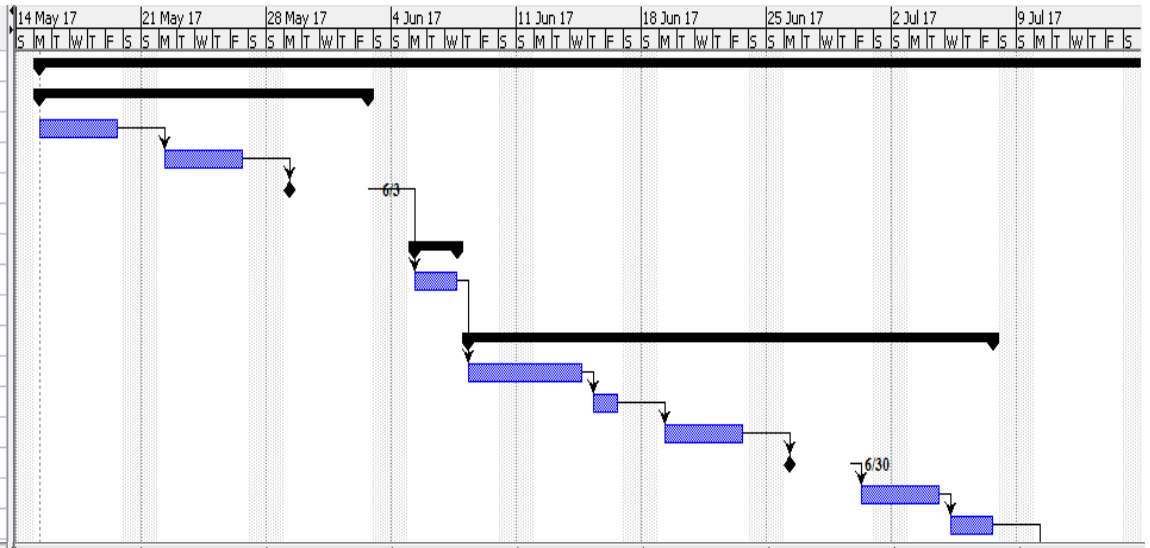
Wong C.K. (April, 2016) 8 facts on mobile commerce growth in Malaysia. Retrieved 1st July 2017, from http://www.ecinsider.my/2016/04/8-facts-on-mobile-commerce-growth-malaysia.html

Xu, L., & Wang, K. (2010, November). Research on influence of chain store logistics capability to logistics service quality. In Logistics Engineering and Intelligent Transportation Systems (LEITS), 2010 International Conference on (pp. 1-4).IEEE.

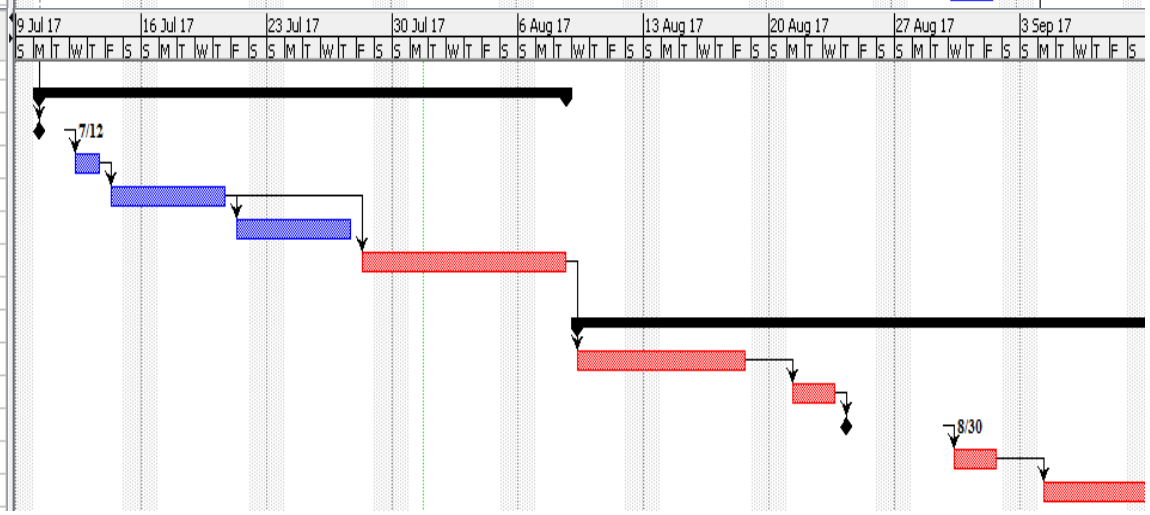
APPENDICES

APPENDIX A: Gantt Chart

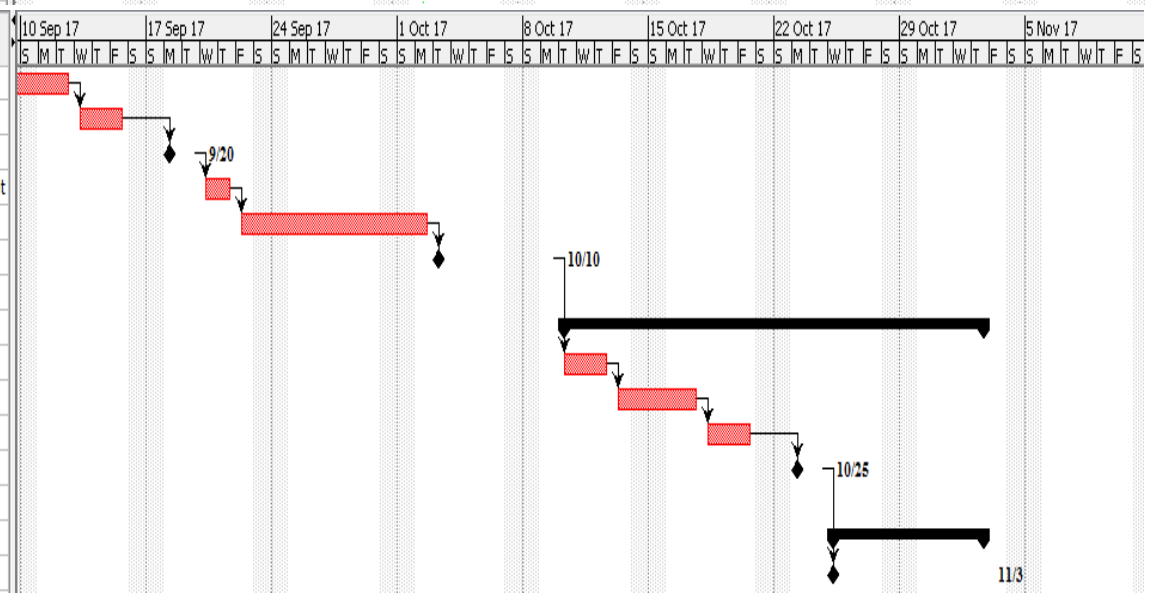
	Name
1	**Parcel2Go Android Application**
2	**Initiation Phase**
3	Project Proposal
4	Prliminary Research
5	Project Approval
6	**Planning Phase**
7	Create Gantt Chart
8	**Analysis Phase**
9	Literature Review Research
10	Compare the existing system with the proposed system
11	Questionnaires distribution
12	Results and Discussions
13	Analyze user needs from survey
14	Analyze system requirement



	Name
15	**Design Phase**
16	UML Diagram Design
17	System Architecture Design
18	User Interface Design
19	Design Database
20	User Interface Coding
21	**Implementation Phase**
22	Build First Prototype : Version 1.0
23	Produce prototype evaluation questionnaire
24	Present the first prototype and questionnaire to user
25	Evaluate the prototype based on functional requirement
26	Refine prototype : Version 2.0



	Name
26	Refine prototype : Version 2.0
27	Documents the adjustments made in the second prototype
28	Present the refine prototype and questionnaire
29	Evaluate the refine prototype based on the functional requirement
30	Final Product : Version 3.0
31	Document adjustment made in the final product
32	**Testing Phase**
33	Identify test requirement
34	Unit Testing
35	Functionality Testing
36	User Acceptance Test
37	**Documentation Phase**
38	Final Documentation of Completion System



APPENDIX B: Preliminary Survey
[GOOGLE SURVEY FORM]

Development of Parcel 2 Go: An Integrated Courier Service Application

HI Everyone! I would like to do a 5 minutes questionnaires survey on my proposed system name Parcel 2 Go : An Integrated Courier Service Application. It helps to integrate a few of the courier services information which includes the price rate and estimated delivery time to reach to the receiver in one stop and offering a low cost peer-to-peer services to people would would like to earn while travelling. Thank you very much for your time and feedback.

Age Group

- 13-17
- 18-24
- 25-34
- 35-44

Occupation

- Student
- Online Business Seller
- Professionals
- Full time Housewife
- Teacher
- Other...

It is very convenient to travel from my place to the nearest post office to send a parcel.

	1	2	3	4	5	
Strongly Disagree	○	○	○	○	○	Strongly Agree

It is very time consuming to run around to get quote from different courier service providers.

	1	2	3	4	5	
Strongly Disagree	○	○	○	○	○	Strongly Agree

It will be very interesting to have a platform to compare few of the courier service providers quotation

	1	2	3	4	5	
Strongly Disagree	○	○	○	○	○	Strongly Agree

It will be better to provide door-to-door delivery service without any registration process needed.

	1	2	3	4	5	
Strongly Disagree	○	○	○	○	○	Strongly Agree

I am willing to earn while travel by helping people to deliver parcels.

	1	2	3	4	5	
Strongly Disagree	○	○	○	○	○	Strongly Agree

Major courier service providers do not provide service to the very rural areas.

	1	2	3	4	5	
Strongly DIsagree	○	○	○	○	○	Strongly Agree

APPENDIX C: User Accepting Testing
Type of User: Customer

Age Group *

- ○ 13-17
- ○ 18-24
- ○ 25-34
- ○ 35-44

Occupations *

- ○ Student
- ○ House-wife
- ○ Professionals
- ○ Online Business Seller
- ○ Other: _____

Parcel2Go can save my time by helping me to get the quotation from different courier services *

	1	2	3	4	5	
Strongly Disagree	○	○	○	○	○	Strongly Agree

Parcel2Go allows me to enjoy the door-to-door delivery service without any terms and conditions apply *

	1	2	3	4	5	
Strongly Disagree	○	○	○	○	○	Strongly Agree

Parcel2Go will ensure that the receiver will safely received the parcel from the personnel *

	1	2	3	4	5	
Strongly Disagree	○	○	○	○	○	Strongly Agree

I will want to use this apps to send my items again *

	1	2	3	4	5	
Strongly Disagree	○	○	○	○	○	Strongly Agree

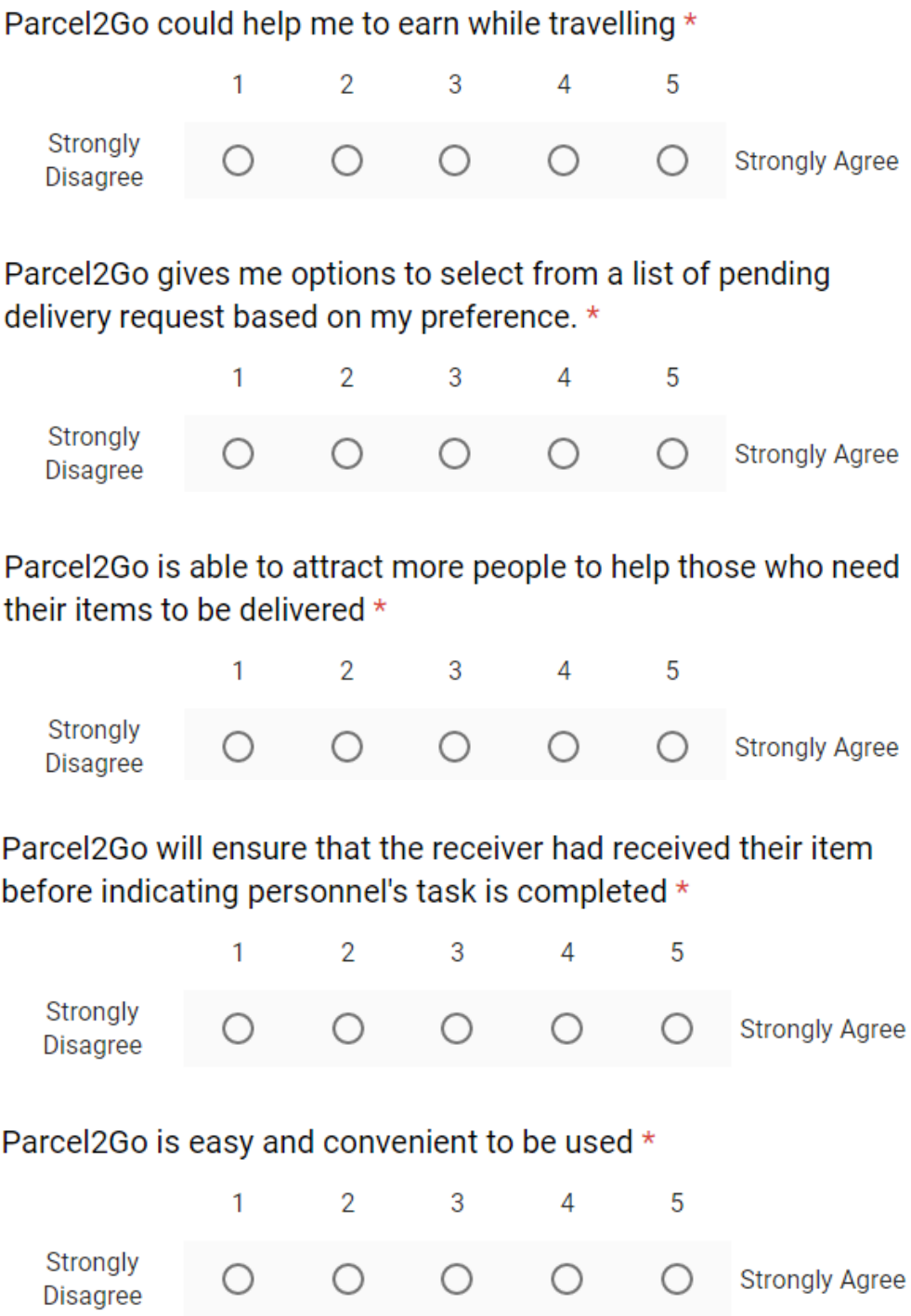
Parcel2Go is easy to use *

	1	2	3	4	5	
Strongly Disagree	○	○	○	○	○	Strongly Agree

It is very convenient to use Parcel2Go to send my items *

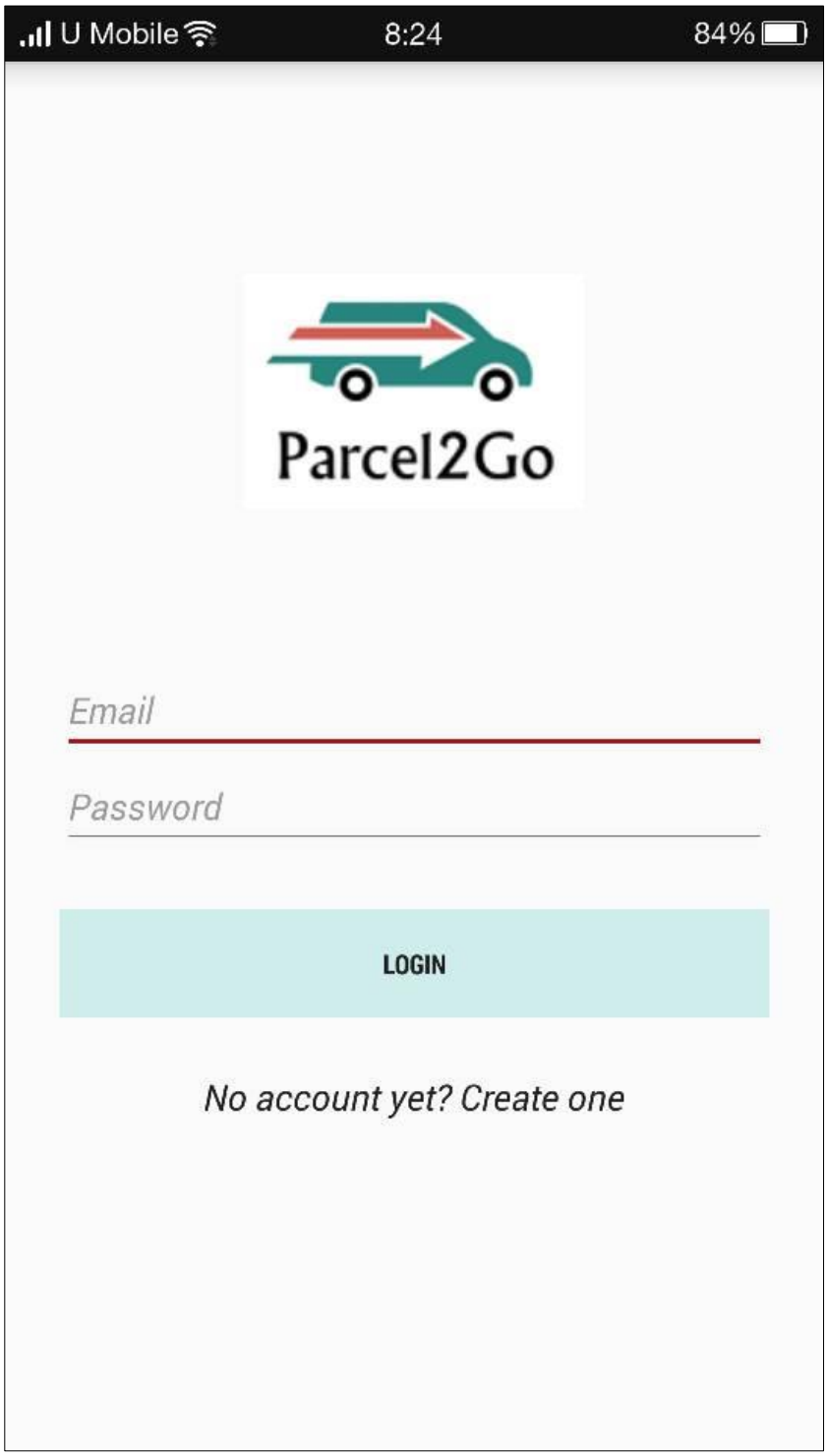
	1	2	3	4	5	
Strongly Disagree	○	○	○	○	○	Strongly Agree

Type of User: Parcel2Go Personnel

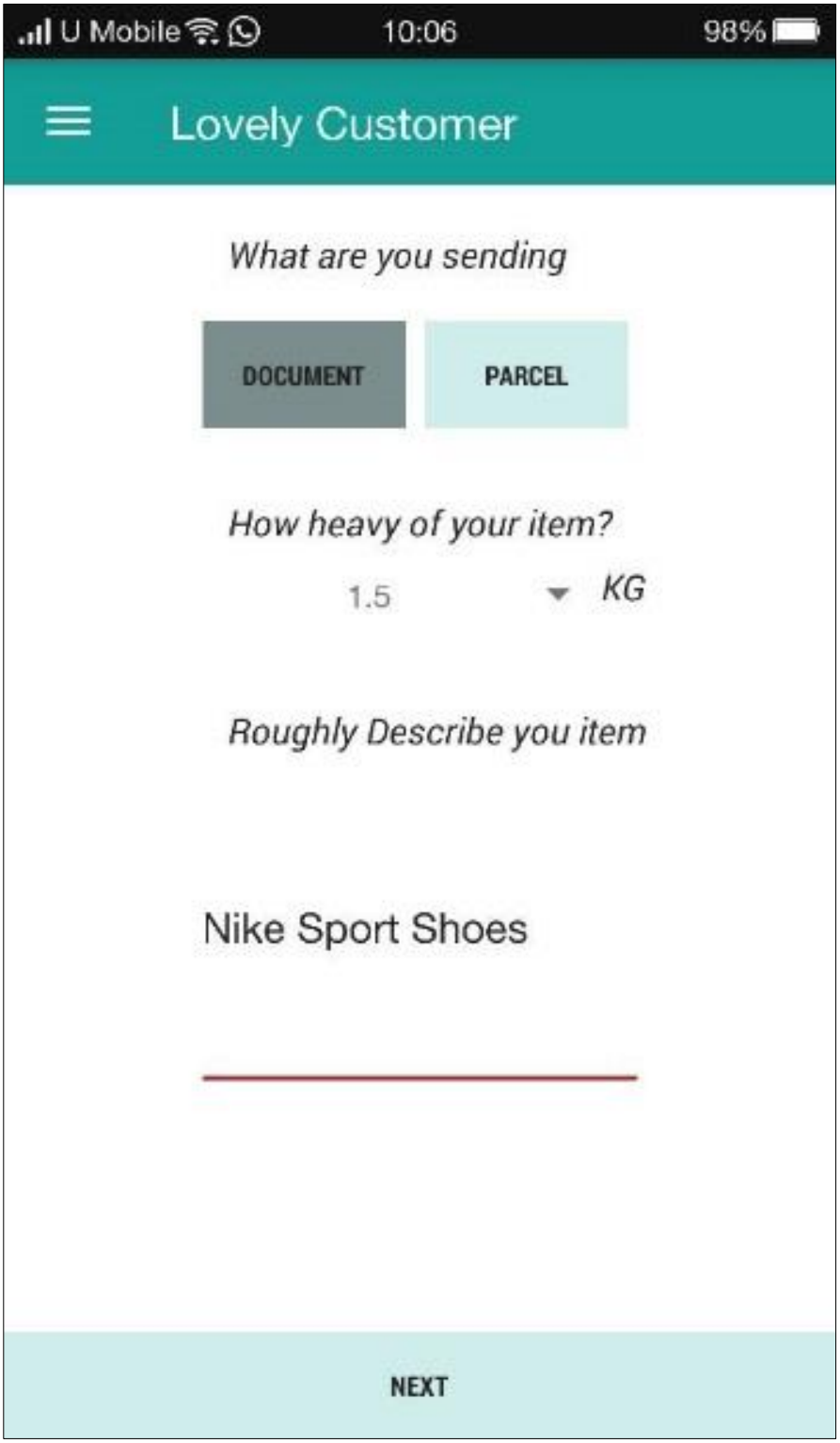


APPENDIX D: Parcel2Go Application Screenshot Interface
CUSTOMER PAGE:

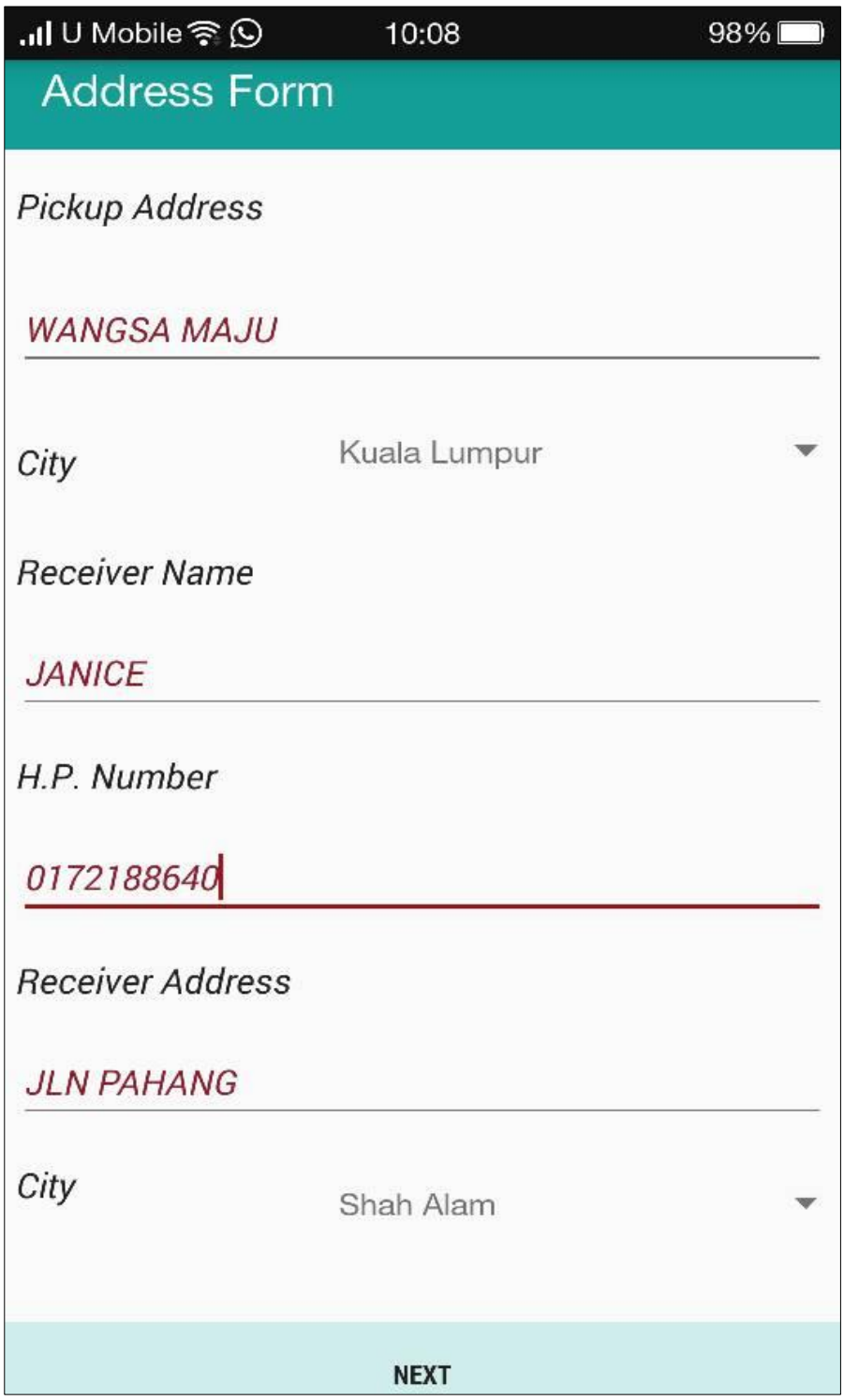
(a) Login Page



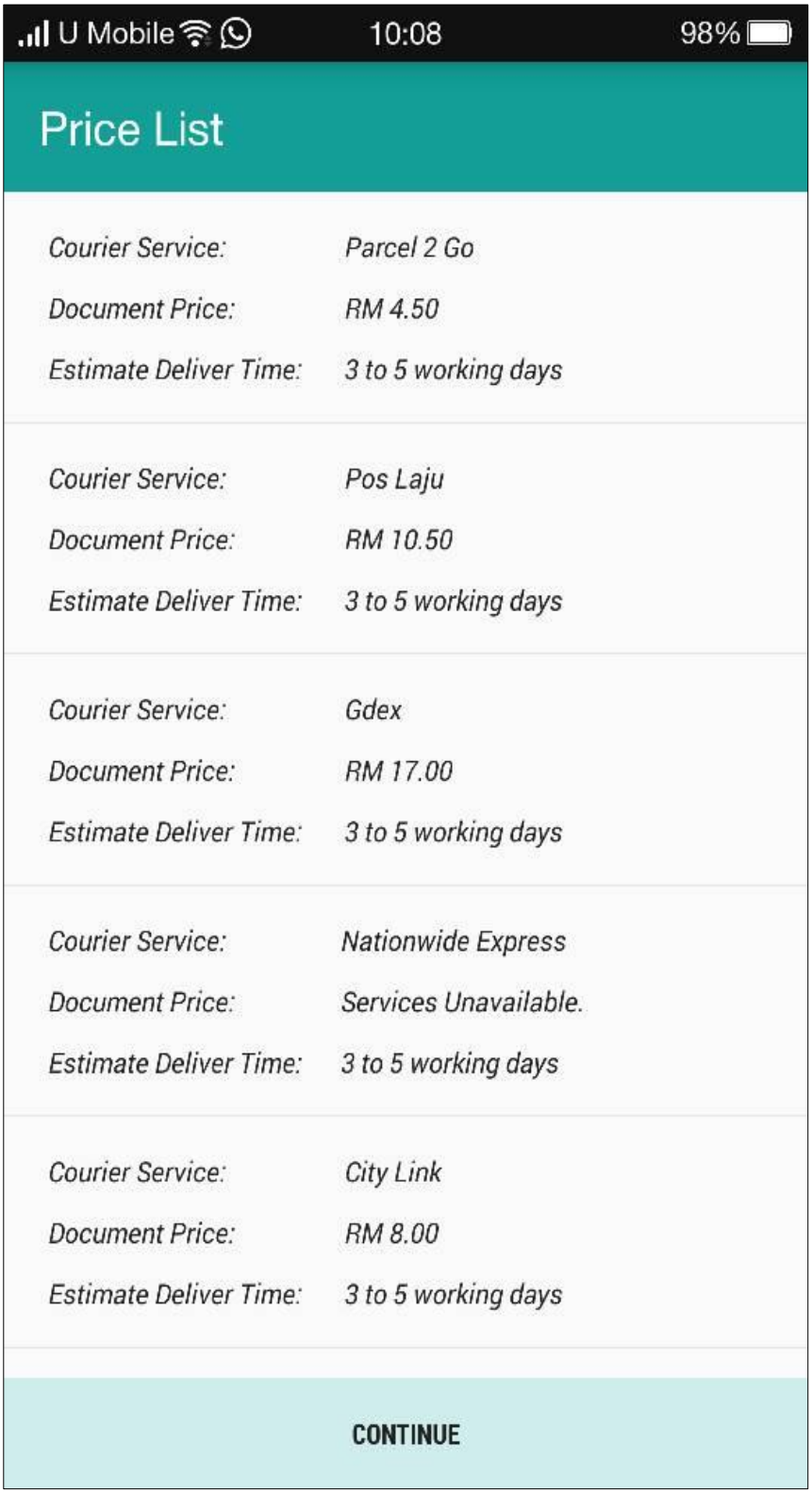
(b) Enter Item Detail Page



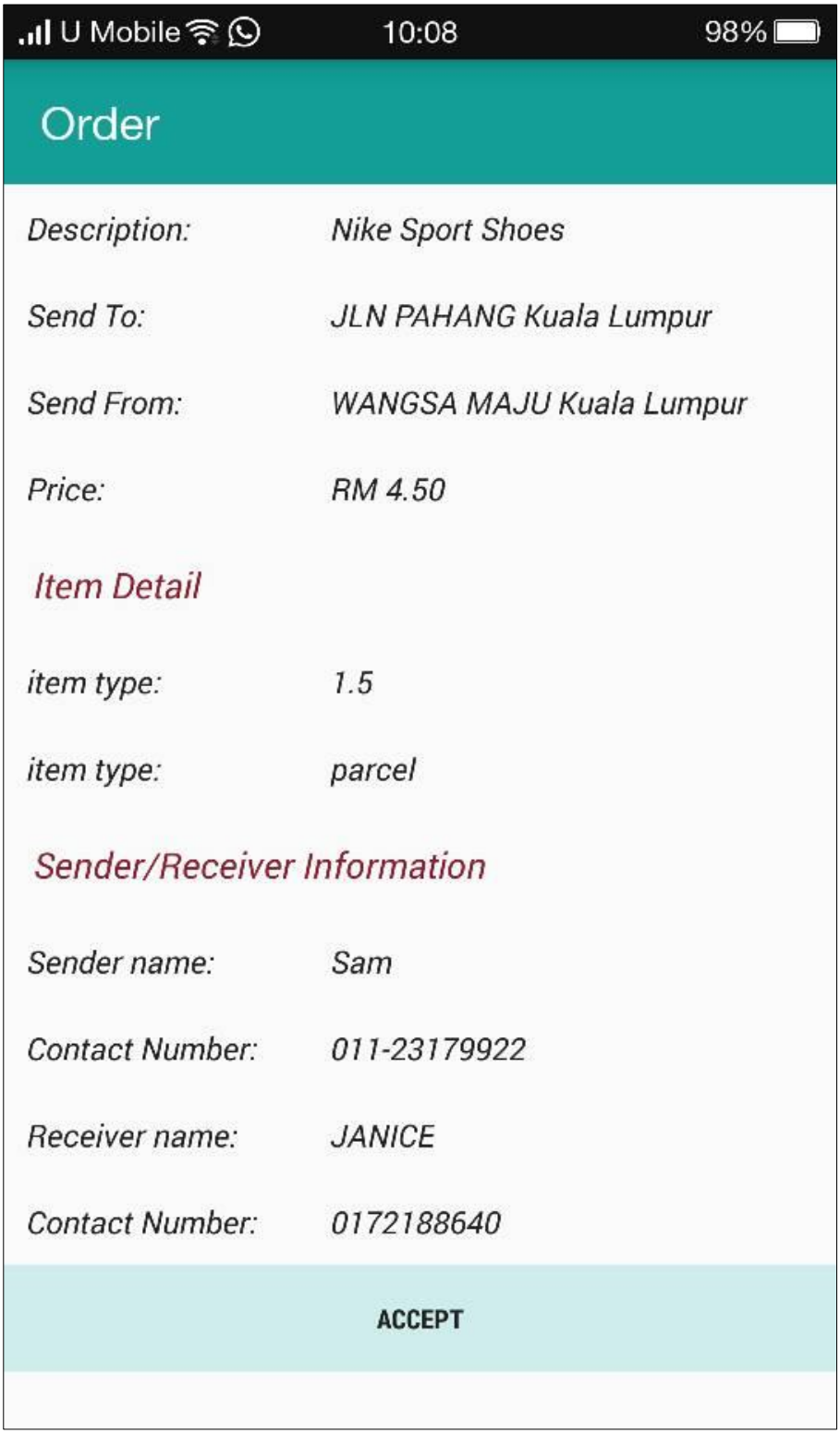
(c) Delivery Destination Page



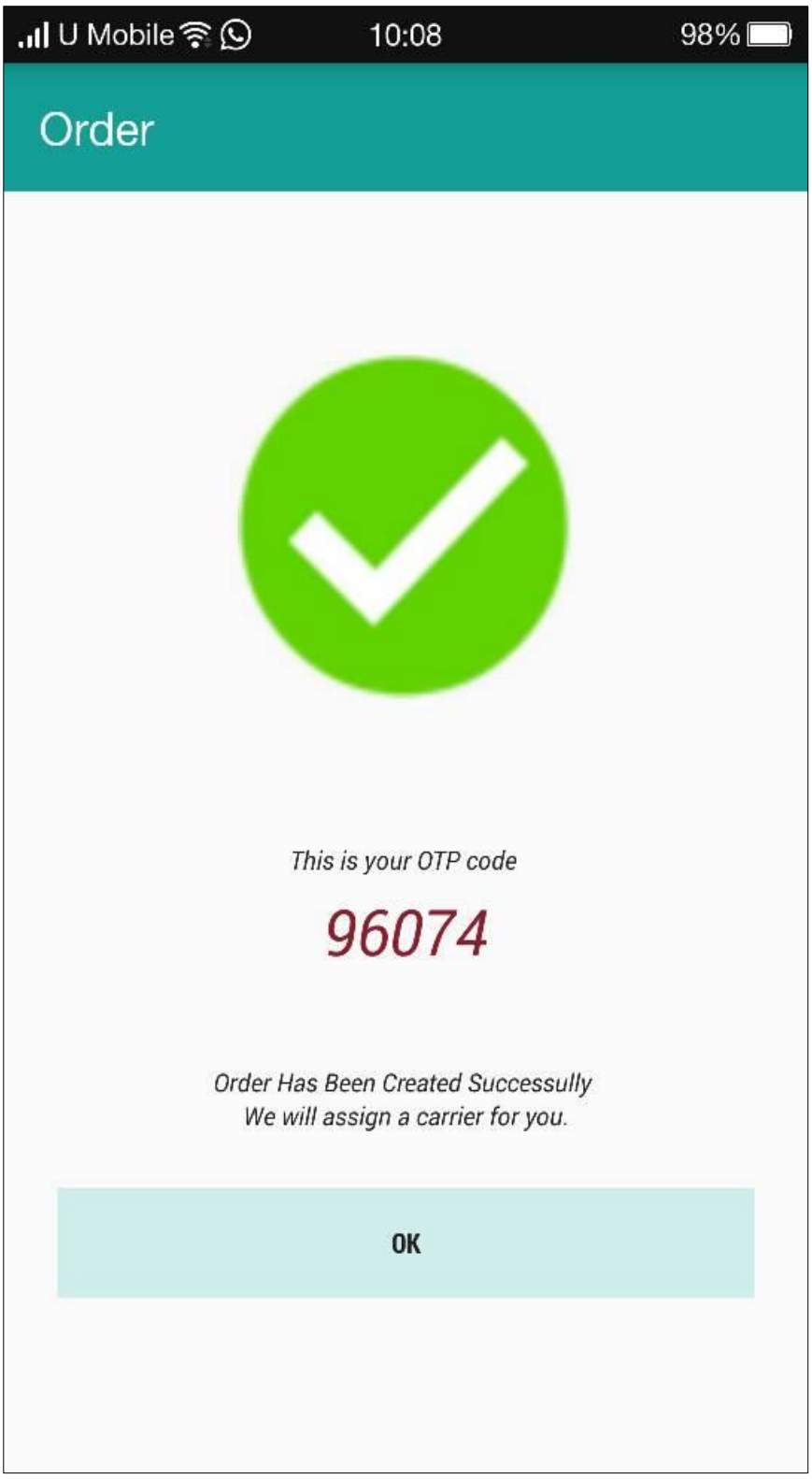
(d) Quote Page


U Mobile
10:08
98%
Price List
Courier Service: Parcel 2 Go
Document Price: RM 4.50
Estimate Deliver Time: 3 to 5 working days
Courier Service: Pos Laju
Document Price: RM 10.50
Estimate Deliver Time: 3 to 5 working days
Courier Service: Gdex
Document Price: RM 17.00
Estimate Deliver Time: 3 to 5 working days
Courier Service: Nationwide Express
Document Price: Services Unavailable.
Estimate Deliver Time: 3 to 5 working days
Courier Service: City Link
Document Price: RM 8.00
Estimate Deliver Time: 3 to 5 working days
CONTINUE

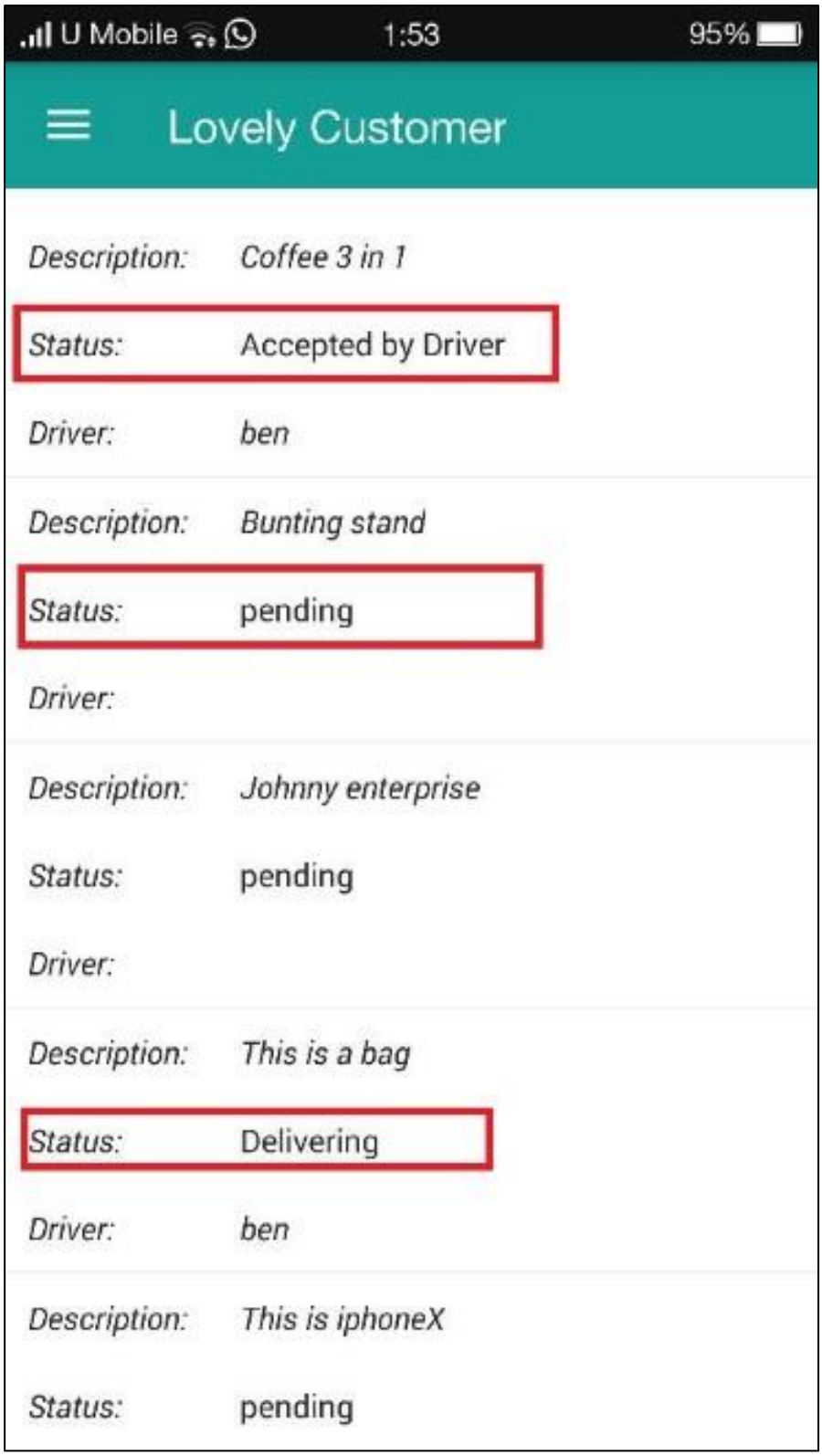
(e) Order Confirmation Page



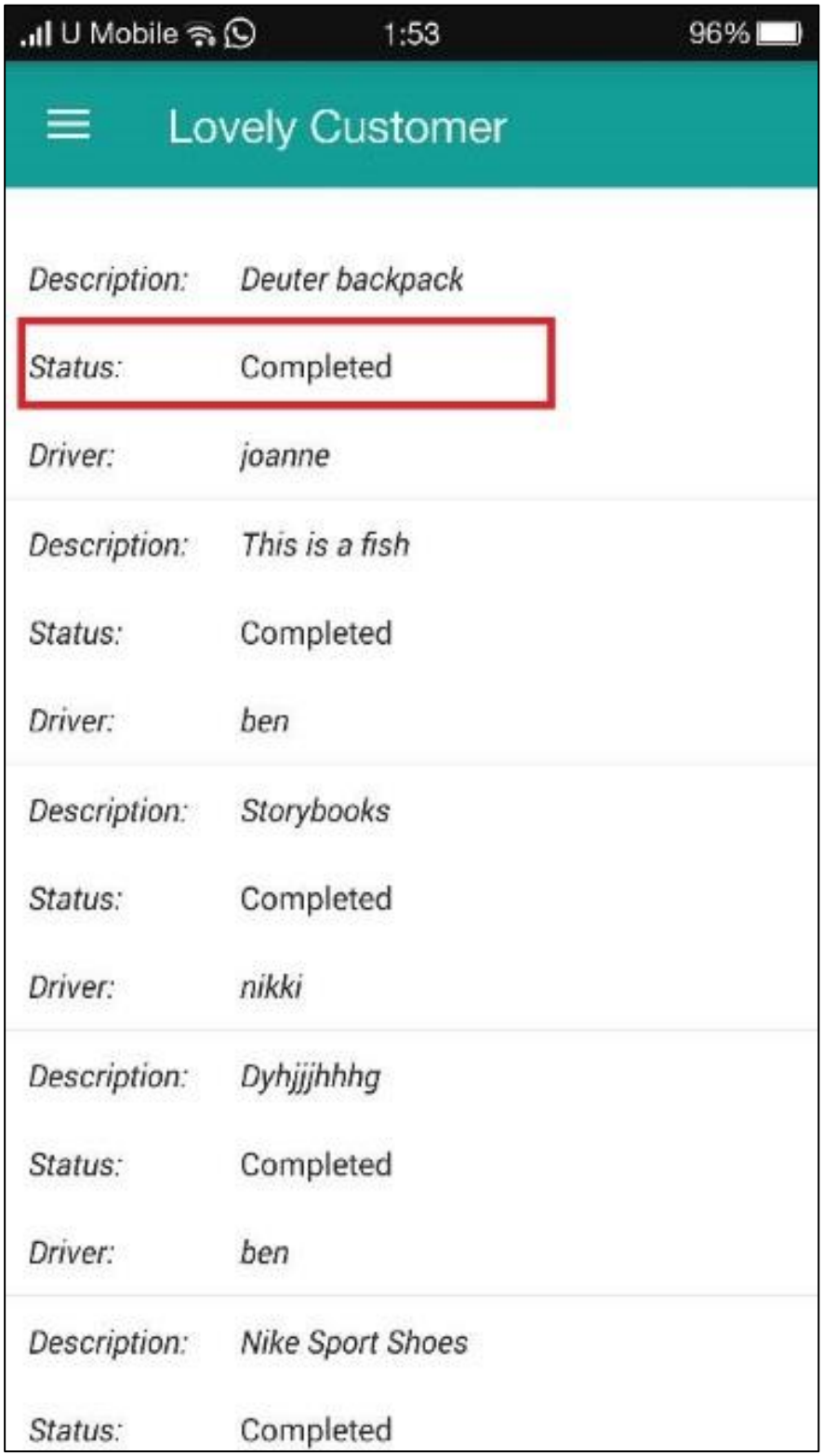
(f) One Time Pass Code Page



(g) Delivery Status Page



(h) Complete Delivery Status Page



PARCEL2GO PERSONNEL PAGE:

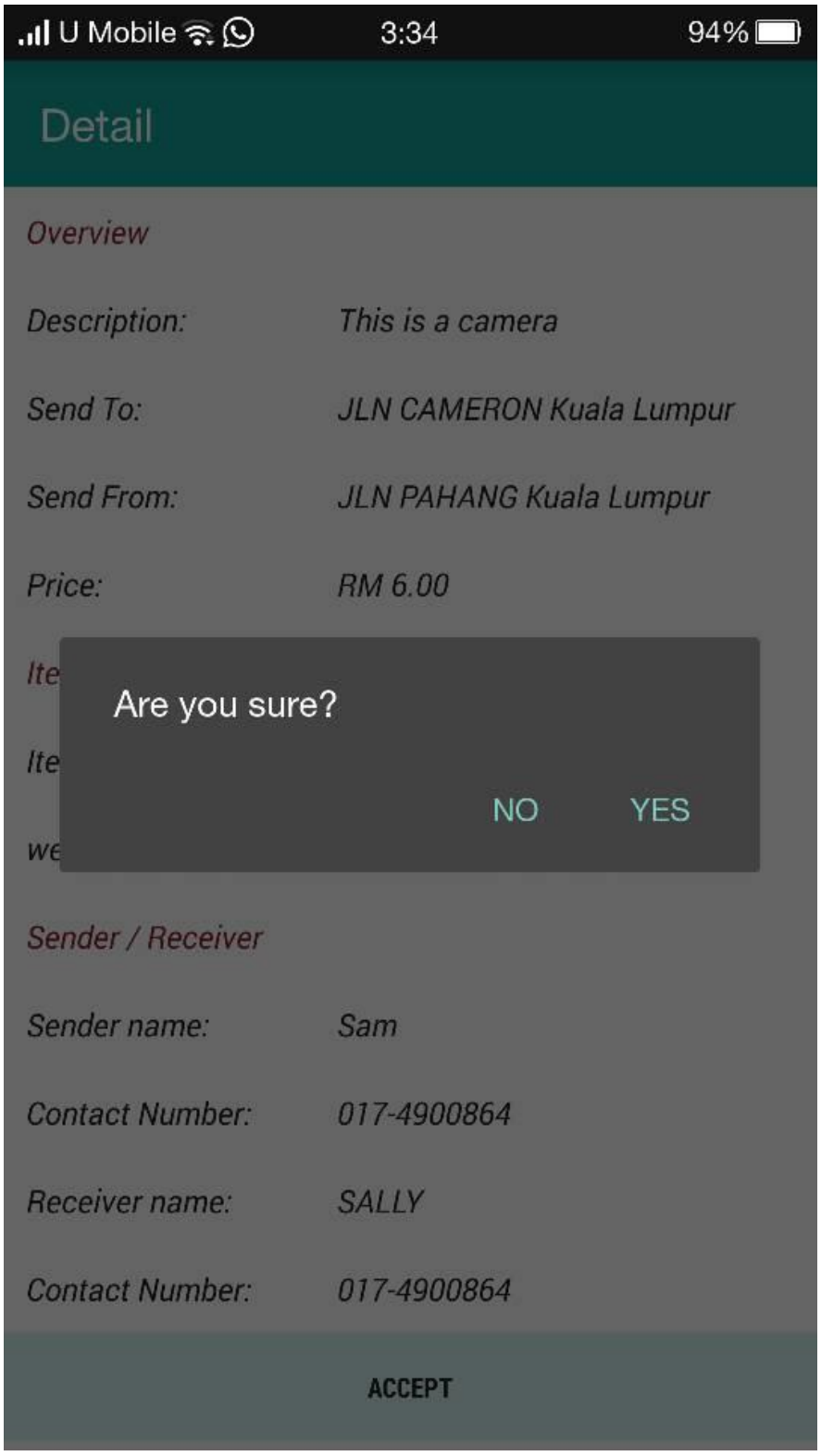
(a) List of Pending Delivery Request



(b) Customer Details Page



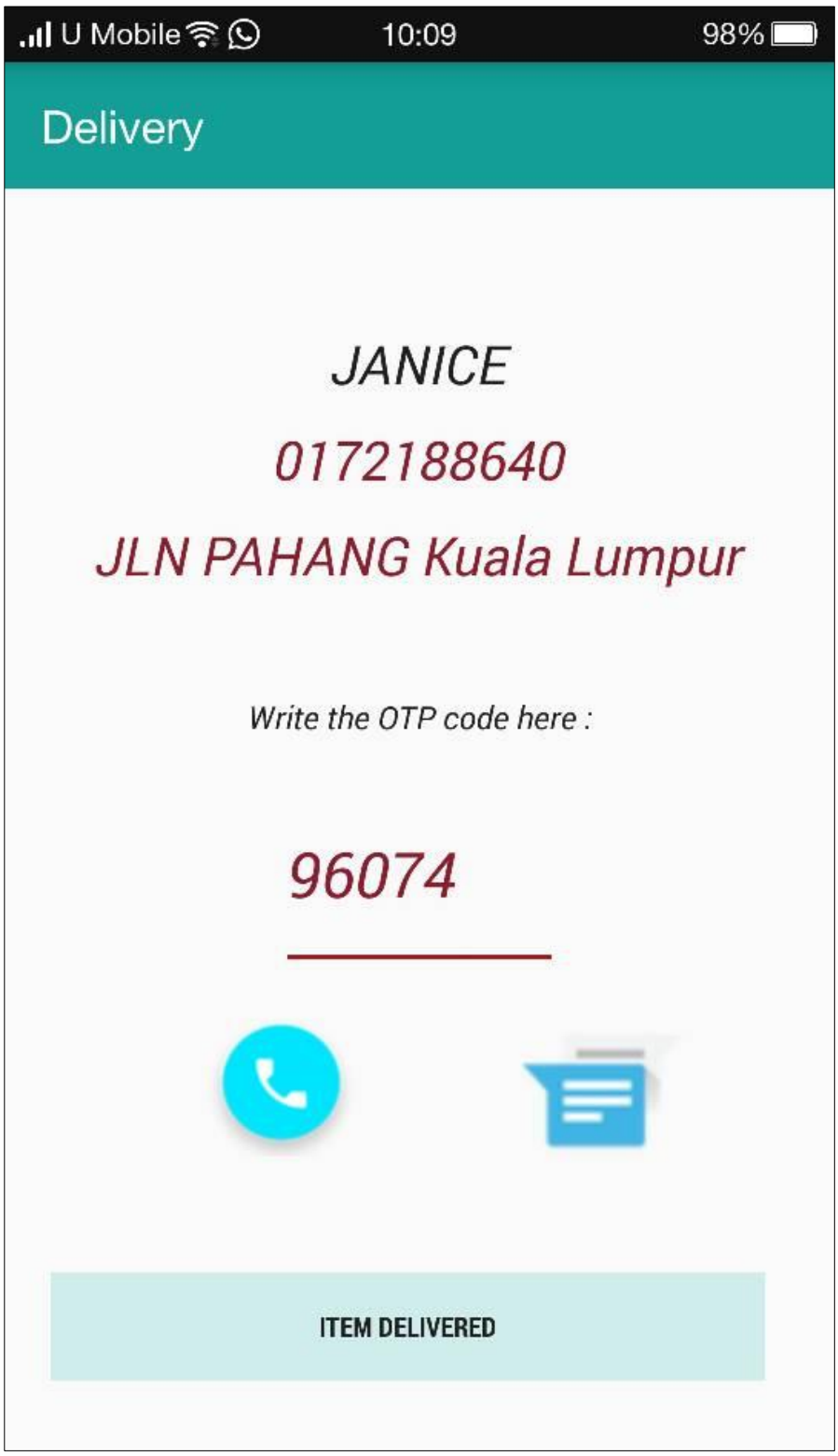
(c) Confirmation Delivery Notification



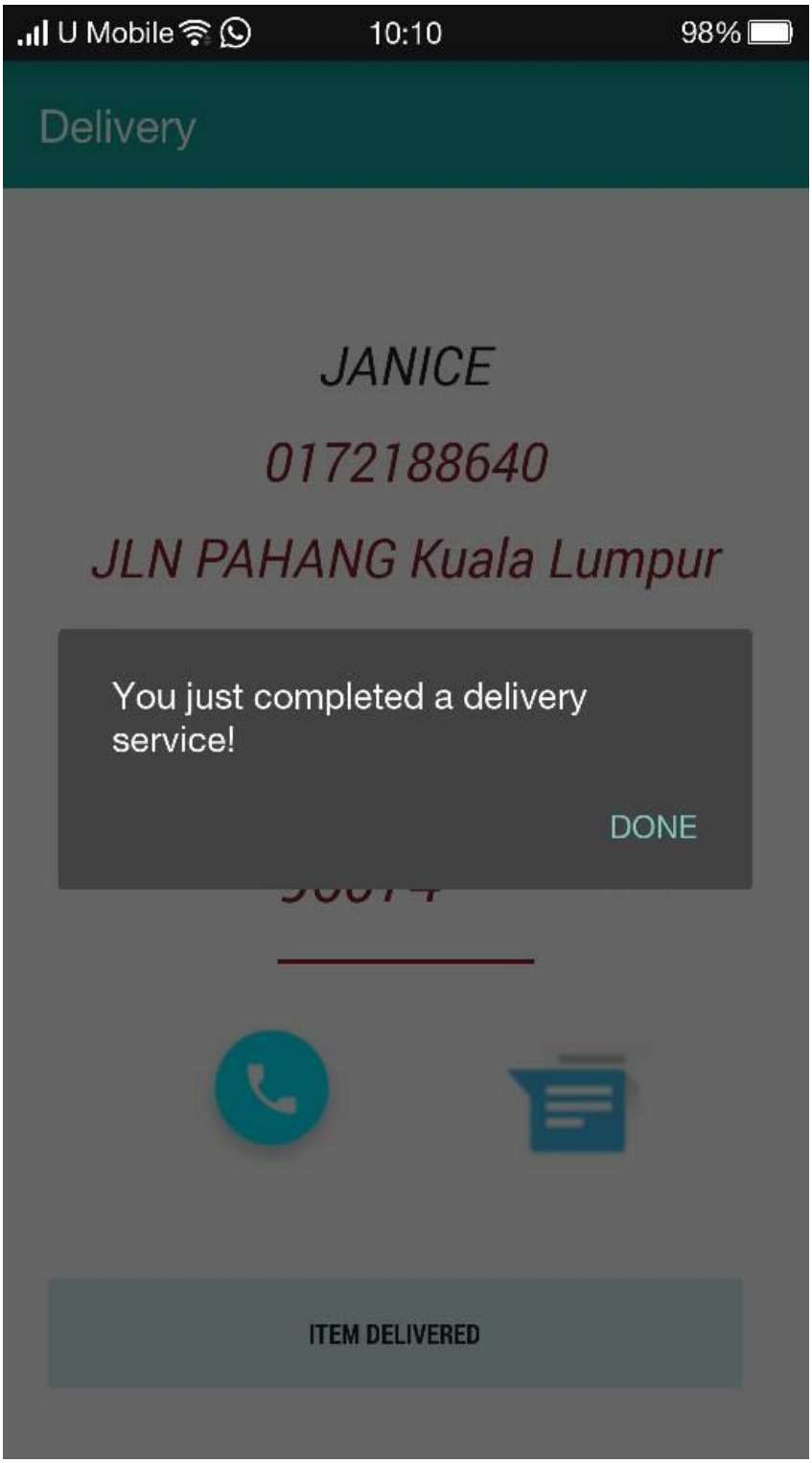
(d) Pick Up Location Page (Sender)



(e) Drop Off Location Page (Receiver)



(f) One Time Pass Code Successfully Verification Page



(g) Driver Profile Page

